

5-9-11

Let f be the function given by $f(x) = 2xe^{2x}$.

1. Find the absolute minimum value of f . Justify that your answer is an absolute minimum.

Part B

1 point is earned if solves $f'(x) = 0$

1 point is earned if evaluates f at student's critical point

0/1 if not local minimum from student's derivative

1 point is earned if justifies absolute minimum value

0/1 for a local argument

0/1 without explicit symbolic derivative

Note: 0/3 if no absolute minimum based on student's derivative

$$f'(x) = 2e^{2x} + 2x \cdot 2 \cdot e^{2x} = 2e^{2x}(1 + 2x) = 0$$

$$\text{if } x = -1/2$$

$$f(-1/2) = -1/e \text{ or } -0.368 \text{ or } -0.367$$

$-1/e$ is an absolute minimum value because:

$$(i) f'(x) < 0 \text{ for all } x < -1/2 \text{ and}$$

$$f'(x) > 0 \text{ for all } x > -1/2$$

$$(ii) f''(x) > 0$$

$$-1/2$$

and $x = -1/2$ is the only critical number



0	1	2	3
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The student response earns three of the following points:

1 point is earned if solves $f'(x) = 0$

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if $x = -1/2$

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$-1/e$ is an absolute minimum value because:

(i) $f'(x) < 0$ for all $x < -1/2$ and

$f'(x) > 0$ for all $x > -1/2$

(ii) $f'(x) - +$

$-1/2$

and $x = -1/2$ is the only critical number

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2. Consider the family of functions defined by $y = bxe^{bx}$, where b is a nonzero constant. Show that the absolute minimum value of bxe^{bx} is the same for all nonzero values of b .

Part D

1 point is earned if sets $y' = be^{bx}(1 + bx) = 0$

1 point is earned for solving student's $y' = 0$

1 point is earned if evaluates y at a critical number and gets a value independent of b

Note: 0/3 if only considering specific values of b

$$y' = be^{bx} + b^2xe^{bx} = be^{bx}(1 + bx) = 0$$

$$\text{if } x = -1/b$$

$$\text{At } x = -1/b, y = -1/e$$

y has an absolute minimum value of $-1/e$ for all nonzero b



0	1	2	3
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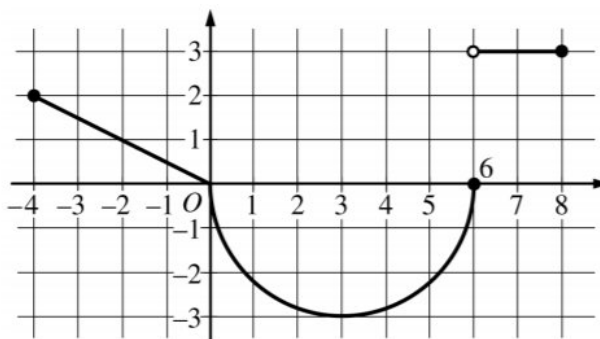
$$\text{if } x = -1/b$$

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y has an absolute minimum value of $-1/e$ for all nonzero b

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3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Graph of g

The function g is defined on the closed interval $[-4, 8]$. The graph of g consists of two linear pieces and a semicircle, as shown in the figure above. Let f be the function defined by $f(x) = 3x + \int_0^x g(t) \, dt$.

(a) Find $f(7)$ and $f'(7)$.

(b) Find the value of x in the closed interval $[-4, 3]$ at which f attains its maximum value. Justify your answer.

(c) For each of $\lim_{x \rightarrow 0^-} g'(x)$ and $\lim_{x \rightarrow 0^+} g'(x)$, find the value or state that it does not exist.

(d) Find $\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1}$.

Part A

1 point is earned for: $f(7)$

1 point is earned for: $f'(7)$

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) \, dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$



0	1	2
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The student response earns all of the following points:

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Part B**2 points are earned for:** answer with justificationOn the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.

0	1	2
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$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} g'(x) \text{ does not exist.}$$



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Part D

1 point is earned for: limits equal 0**1 point is earned for:** applies L'Hospital's Rule**1 point is earned for:** answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives



0	1	2	3
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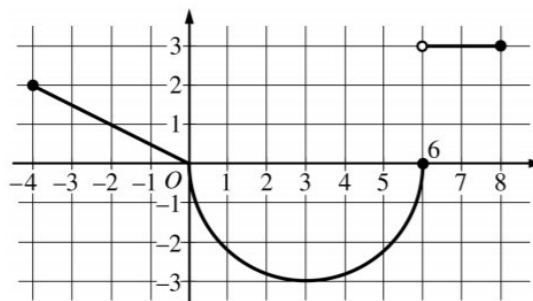
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(a) Find $f(7)$ and $f'(7)$.

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0	1	2
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Part D

1 point is earned for: limits equal 0**1 point is earned for:** applies L'Hospital's Rule**1 point is earned for:** answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives



0	1	2	3
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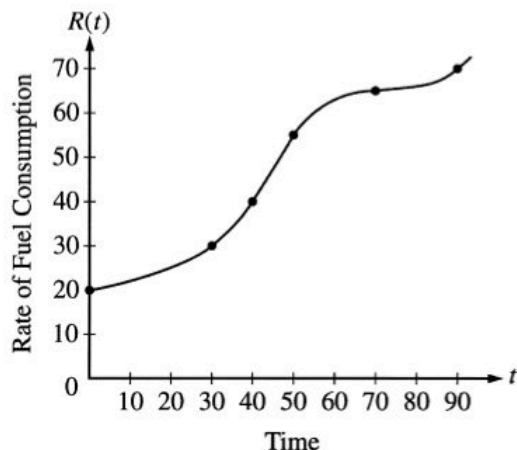
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Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

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5.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

The rate of fuel consumption is increasing fastest at time $t = 45$ minutes. What is the value of $R''(45)$? Explain your reasoning.

Part B

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.



0	1	2
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The student response earns two of the following points:

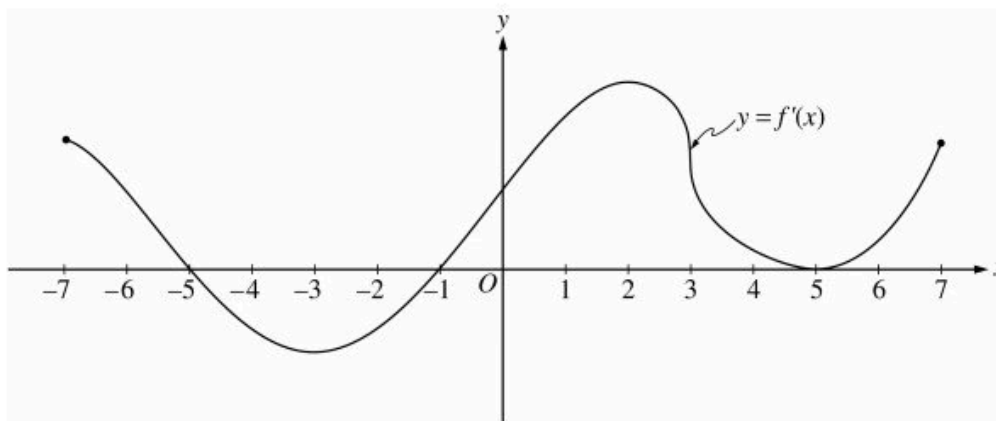
1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.

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6.



The figure above shows the graph of f' , the derivative of the function f , for $-7 \leq x \leq 7$. The graph of f has horizontal tangent lines at $x = -3$, $x = 2$, and $x = 5$, and a vertical tangent line at $x = 3$.

- Find all values of x , for $-7 < x < 7$, at which f attains a relative minimum. Justify your answer.
- Find all values of x , for $-7 < x < 7$, at which f attains a relative maximum. Justify your answer.
- Find all values of x , for $-7 < x < 7$, at which $f''(x) < 0$.
- At what value of x , for $-7 \leq x \leq 7$, does f attain its absolute maximum? Justify your answer.

Part A

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$ changes from negative to positive at $x = -1$



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$ changes from negative to positive at $x = -1$

Part B

1 point is earned for the answer

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1 point is earned for the justification

$$x = -5$$

$f'(x)$ changes from positive to negative at $x = -5$



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$x = -5$$

$f'(x)$ changes from positive to negative at $x = -5$

Part C

1 point is earned for $(-7, -3)$

1 point is earned for $(2, 3) \cup (3, 5)$

$f''(x)$ exists and f' is decreasing on the intervals $(-7, -3)$, $(2, 3)$, and $(3, 5)$



0	1	2
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Part D

1 point is earned for the answer

1 point is earned for identifying $x = -5$ and $x = 7$ as candidates

- or -

indicating that the graph of f increases, decreases, then increases

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1 point is earned for justifying $f(7) > f(-5)$

$$x = 7$$

The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.



0	1	2	3
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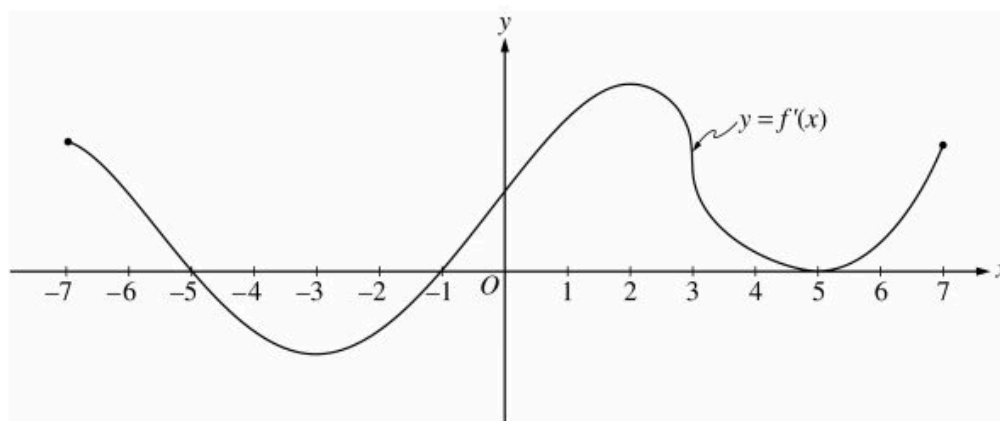
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7.



The figure above shows the graph of f' , the derivative of the function f , for $-7 \leq x \leq 7$. The graph of f has horizontal tangent lines at $x = -3$, $x = 2$, and $x = 5$, and a vertical tangent line at $x = 3$.

Find all values of x , for $-7 < x < 7$, at which $f''(x) < 0$.

Part C

1 point is earned for: $(-7, -3)$

1 point is earned for: $(2, 3) \cup (3, 5)$

$f''(x)$ exists and f' is decreasing on the intervals

$(-7, -3)$, $(2, 3)$, and $(3, 5)$

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0	1	2
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The student response earns all of the following points:

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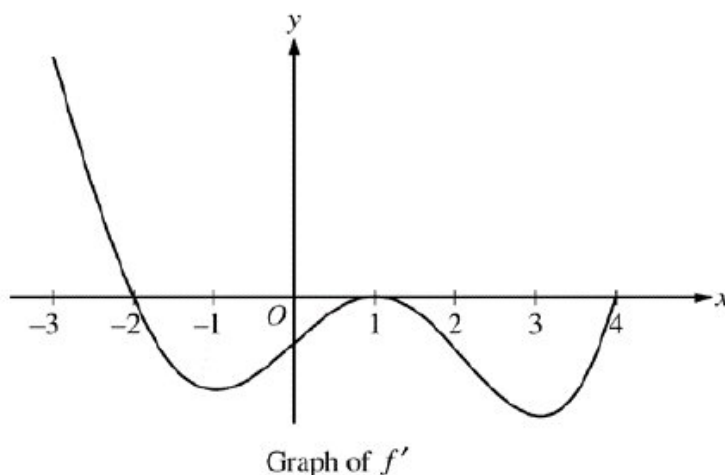
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$f''(x)$ exists and f' is decreasing on the intervals

$(-7, -3)$, $(2, 3)$, and $(3, 5)$

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8.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the interval $[-3, 4]$. The graph of f' has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-2, 1]$ and $[1, 4]$ are 9 and 12, respectively.

- Find all x -coordinates at which f has a relative maximum. Give a reason for your answer.
- On what open intervals contained in $-3 < x < 4$ is the graph of f both concave down and decreasing? Give a reason for your answer.
- Find the x -coordinates of all points of inflection for the graph of f . Give a reason for your answer.
- Given that $f(1) = 3$, write an expression for $f(x)$ that involves an integral. Find $f(4)$ and $f(-2)$.

Part A

1 point is earned for identifying $x = -2$

1 point is earned for answer with reason

$f'(x) = 0$ at $x = -2$, $x = 1$, and $x = 4$.

$f'(x)$ changes from positive to negative at $x = -2$.

Therefore, f has a relative maximum at $x = -2$.



0	1	2
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The student response earns all of the following points:

1 point is earned for identifying $x = -2$

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$f'(x) = 0$ at $x = -2$, $x = 1$, and $x = 4$.

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$f(x)$ changes from positive to negative at $x = -2$.

Therefore, f has a relative maximum at $x = -2$.

Part B

1 point is earned for the intervals

1 point is earned for the reason

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.



0	1	2
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The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the reason

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.

Part C

1 point is earned for identifying $x = -1, 1$, and 3

1 point is earned for the reason

The graph of f has a point of inflection at $x = -1$ and $x = 3$ because f' changes from decreasing to increasing at these points. The graph of f has a point of inflection at $x = 1$ because f' changes from increasing to decreasing at this point.



0	1	2
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1 point is earned for identifying $x = -1, 1$, and 3

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The graph of f has a point of inflection at $x = -1$ and $x = 3$ because f' changes from decreasing to increasing at these points. The graph of f has a point of inflection at $x = 1$ because f' changes from increasing to decreasing at this point.

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Part D

1 point is earned for the integrand

1 point is earned for expressing $f(x)$ 1 point is earned for $f(4)$ and $f(-2)$

$$f(x) = 3 + \int_1^x f'(t) dt$$

$$f(4) = 3 + \int_1^4 f'(t) dt = 3 + (-12) = -9$$

$$\begin{aligned} f(-2) &= 3 + \int_1^{-2} f'(t) dt = 3 - \int_{-2}^1 f'(t) dt \\ &= 3 - (-9) = 12 \end{aligned}$$



0	1	2	3
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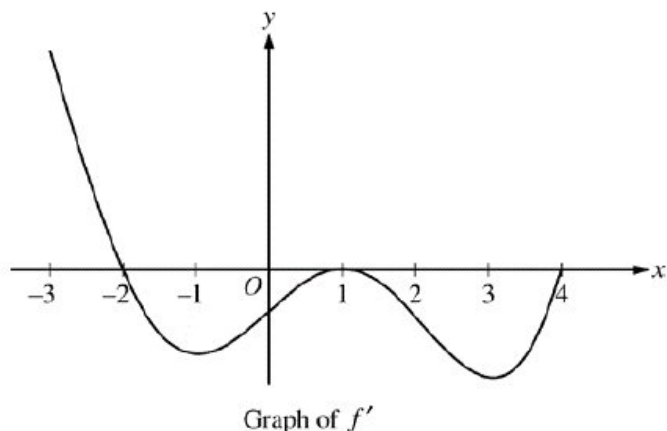
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9.



Find the x -coordinates of all points of inflection for the graph of f . Give a reason for your answer.

Part C

1 point is earned for identifying $x = -1, 1$, and 3

1 point is earned for the reason

The graph of f has a point of inflection at $x = -1$ and $x = 3$ because f' changes from decreasing to increasing at these points. The graph of f has a point of inflection at $x = 1$ because f' changes from increasing to decreasing at this point.



0	1	2
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1 point is earned for identifying $x = -1, 1$, and 3

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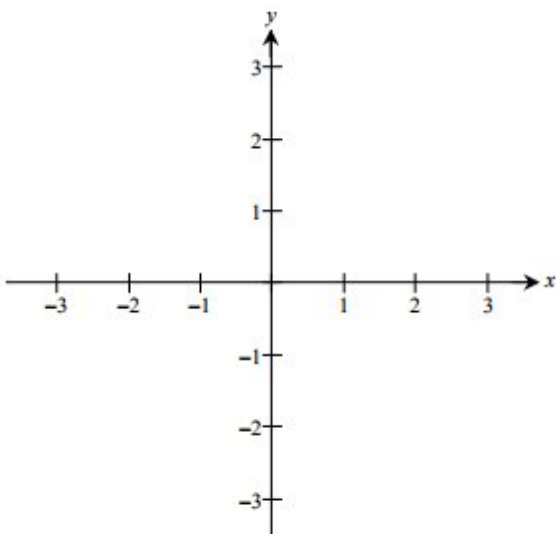
The graph of f has a point of inflection at $x = 1$ because f' changes from increasing to decreasing at this point.

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10. Let f be a function that is even and continuous on the closed interval $[-3, 3]$. The function f and its derivatives have the properties indicated in the table below.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$
$f(x)$	1	Positive	0	Negative	-1	Negative
$f'(x)$	Undefined	Negative	0	Negative	Undefined	Positive
$f''(x)$	Undefined	Positive	0	Negative	Undefined	Negative

In the xy -plane provided below, sketch the graph of a function with all the given characteristics of f .

**Part C**

3 point is earned for sketch without error

1 point is deducted for each error:

symmetry

slopes on $[0, 3)$

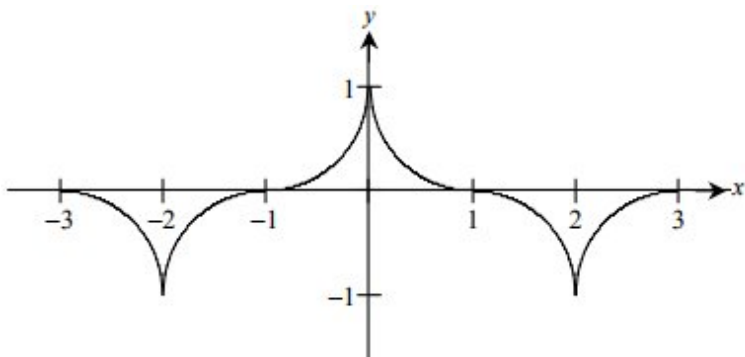
concavity on $[0, 3)$

discontinuity on $[0, 3)$

$f(0), f(1), f(2)$ incorrect or extra intercept on $[0, 3)$

Note: No more than 1 point can be lost in any one category.

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0	1	2	3
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The student response earns three of the following points:

3 point is earned for sketch without error

1 point is deducted for each error:

symmetry

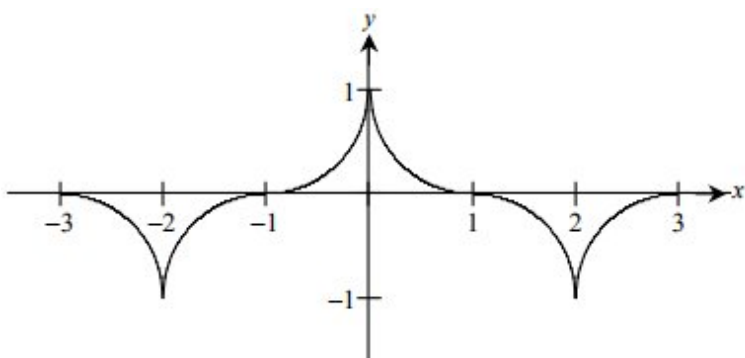
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11. Let $g(x) = xe^{-x} + be^{-x}$, where b is a positive constant.

(a) Find $\lim_{x \rightarrow \infty} g(x)$.

(b) For what positive value of b does g have an absolute maximum at $x = \frac{2}{3}$? Justify your answer.

(c) Find all values of b , if any, for which the graph of g has a point of inflection on the interval $0 < x < \infty$. Justify your answer.

Part A

1 point is earned for the answer

$$\lim_{x \rightarrow \infty} g(x) = 0$$



0	1
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The student response earns all of the following points:

1 point is earned for the answer

$$\lim_{x \rightarrow \infty} g(x) = 0$$

Part B

2 points are earned for $g'(x)$

1 point is earned for solving $g'\left(\frac{2}{3}\right) = 0$ for b

1 point is earned for justification

$$g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$$

$$g'\left(\frac{2}{3}\right) = \left(\frac{1}{3} - b\right)e^{-2/3} = 0 \Rightarrow b = \frac{1}{3}$$

$$\text{When } b = \frac{1}{3}, g'(x) = \left(\frac{2}{3} - x\right)e^{-x}.$$

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For $x < \frac{2}{3}$, $g'(x) > 0$ and for $x > \frac{2}{3}$, $g'(x) < 0$.

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum at $x = \frac{2}{3}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for solving $g'\left(\frac{2}{3}\right) = 0$ for b

1 point is earned for justification

$$g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$$

$$g'\left(\frac{2}{3}\right) = \left(\frac{1}{3} - b\right)e^{-2/3} = 0 \Rightarrow b = \frac{1}{3}$$

$$\text{When } b = \frac{1}{3}, g'(x) = \left(\frac{2}{3} - x\right)e^{-x}.$$

For $x < \frac{2}{3}$, $g'(x) > 0$ and for $x > \frac{2}{3}$, $g'(x) < 0$.

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum at $x = \frac{2}{3}$.

Part C

2 points are earned for $g''(x)$

1 point is earned for interval for b

1 point is earned for justification

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$$g''(x) = -e^{-x} - (1 - x - b)e^{-x} = (x - 2 + b)e^{-x}$$

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$.

Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $g''(x)$

1 point is earned for interval for b

1 point is earned for justification

$$g''(x) = -e^{-x} - (1 - x - b)e^{-x} = (x - 2 + b)e^{-x}$$

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$.

Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.

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12.  Let f be the function given by $f(x) = 2xe^{2x}$.

(a) Find $\lim_{x \rightarrow -\infty} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$.

(b) Find the absolute minimum value of f . Justify that your answer is an absolute minimum.

(c) What is the range of f ?

(d) Consider the family of functions defined by $y = bxe^{bx}$, where b is a nonzero constant. Show that the absolute minimum value of bxe^{bx} is the same for all nonzero values of b .

Part A

1 point is earned for 0 as $x \rightarrow -\infty$

1 point is earned for ∞ or DNE as $x \rightarrow \infty$

$$\lim_{x \rightarrow -\infty} 2xe^{2x} = 0$$

$$\lim_{x \rightarrow \infty} 2xe^{2x} = \infty \text{ or DNE}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for 0 as $x \rightarrow -\infty$

1 point is earned for ∞ or DNE as $x \rightarrow \infty$

$$\lim_{x \rightarrow -\infty} 2xe^{2x} = 0$$

$$\lim_{x \rightarrow \infty} 2xe^{2x} = \infty \text{ or DNE}$$

Part B

1 point is earned if solves $f'(x) = 0$

1 point is earned if evaluates f at student's critical point

0/1 if not local minimum from student's derivative

1 point is earned if justifies absolute minimum value

0/1 for a local argument

0/1 without explicit symbolic derivative

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Note: 0/3 if no absolute minimum based on student's derivative

$$f'(x) = 2e^{2x} + 2x \cdot 2 \cdot e^{2x} = 2e^{2x}(1 + 2x) = 0$$

if $x = -1/2$

$$f(-1/2) = -1/e \text{ or } -0.368 \text{ or } -0.367$$

$-1/e$ is an absolute minimum value because:

(i) $f'(x) < 0$ for all $x < -1/2$ and $f'(x) > 0$ for all $x > -1/2$

-or-

(ii)
$$\begin{array}{c} f'(x) \quad - \quad + \\ \hline -1/2 \end{array}$$

and $x = -1/2$ is the only critical number



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if solves $f'(x) = 0$

1 point is earned if evaluates f at student's critical point

0/1 if not local minimum from student's derivative

1 point is earned if justifies absolute minimum value

0/1 for a local argument

0/1 without explicit symbolic derivative

Note: 0/3 if no absolute minimum based on student's derivative

$$f'(x) = 2e^{2x} + 2x \cdot 2 \cdot e^{2x} = 2e^{2x}(1 + 2x) = 0$$

if $x = -1/2$

$$f(-1/2) = -1/e \text{ or } -0.368 \text{ or } -0.367$$

$-1/e$ is an absolute minimum value because:

(i) $f'(x) < 0$ for all $x < -1/2$ and $f'(x) > 0$ for all $x > -1/2$

-or-

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(ii) $f'(x)$ $\frac{-}{+}$
 $\frac{-1/2}{-1/2}$

and $x = -1/2$ is the only critical number

Part C

1 point is earned for the answer

Note: must include the left-hand endpoint; exclude the right-hand “endpoint”

Range of $f = [-1/e, \infty)$ or $[-0.367, \infty)$ or $[-0.368, \infty)$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

Note: must include the left-hand endpoint; exclude the right-hand “endpoint”

Range of $f = [-1/e, \infty)$ or $[-0.367, \infty)$ or $[-0.368, \infty)$

Part D

1 point is earned if sets $y' = be^{bx}(1 + bx) = 0$

1 point is earned for solving student's $y' = 0$

1 point is earned if evaluates y at a critical number and gets a value independent of b

Note: 0/3 if only considering specific values of b

$$y' = be^{bx} + b^2xe^{bx} = be^{bx}(1 + bx) = 0$$

if $x = -1/b$

At $x = -1/b, y = -1/e$

y has an absolute minimum value of $-1/e$ for all nonzero b



0	1	2	3
---	---	---	---

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The student response earns all of the following points:

1 point is earned if sets $y' = be^{bx}(1 + bx) = 0$

1 point is earned for solving student's $y' = 0$

1 point is earned if evaluates y at a critical number and gets a value independent of b

Note: 0/3 if only considering specific values of b

$$y' = be^{bx} + b^2xe^{bx} = be^{bx}(1 + bx) = 0$$

$$\text{if } x = -1/b$$

$$\text{At } x = -1/b, y = -1/e$$

y has an absolute minimum value of $-1/e$ for all nonzero b

5-9-11

13. Consider the function f given by $f(x) = xe^{-x^2}$ for all real numbers x .
At what value of x does $f(x)$ attain its absolute maximum? Justify your answer.

Part A

Two points are earned for $f'(x)$

One point is earned for $f'(x) = 0$

One point is earned for the answer

One point is earned for the justification

$$f'(x) = e^{-x^2} - 2x^2e^{-x^2} = (1 - 2x^2)e^{-x^2}$$

$$f'(x) = 0 \text{ when } x = -\frac{1}{\sqrt{2}} \text{ or } x = \frac{1}{\sqrt{2}}$$

$f(x) < 0$ for $x < 0$, and $f(x) > 0$ for $x > 0$. Therefore, a maximum can only occur at a positive x -value. $f'(x)$

changes sign from positive to negative at $x = \frac{1}{\sqrt{2}}$, so f has a relative maximum at $x = \frac{1}{\sqrt{2}}$. f increases on $0 < x$

$\leq \frac{1}{\sqrt{2}}$ and decreases on $\frac{1}{\sqrt{2}} \leq x < \infty$. Therefore, there is an absolute maximum at $x = \frac{1}{\sqrt{2}}$.



0	1	2	3	4	5
---	---	---	---	---	---

The student earns all of the following points:

Two points are earned for $f'(x)$

One point is earned for $f'(x) = 0$

One point is earned for the answer

One point is earned for the justification

$$f'(x) = e^{-x^2} - 2x^2e^{-x^2} = (1 - 2x^2)e^{-x^2}$$

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$$f'(x) = 0 \text{ when } x = -\frac{1}{\sqrt{2}} \text{ or } x = \frac{1}{\sqrt{2}}$$

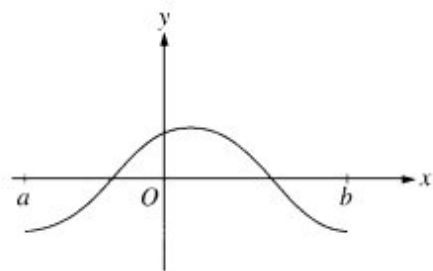
$f(x) < 0$ for $x < 0$, and $f(x) > 0$ for $x > 0$. Therefore, a maximum can only occur at a positive x -value. $f'(x)$

changes sign from positive to negative at $x = \frac{1}{\sqrt{2}}$, so f has a relative maximum at $x = \frac{1}{\sqrt{2}}$. f increases on $0 < x$

$< \frac{1}{\sqrt{2}}$ and decreases on $\frac{1}{\sqrt{2}} \leq x < \infty$. Therefore, there is an absolute maximum at $x = \frac{1}{\sqrt{2}}$.

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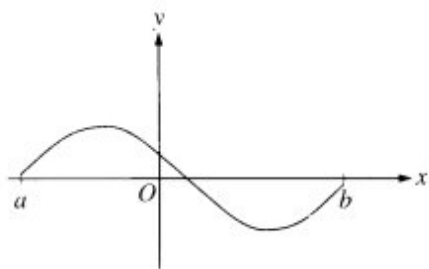
14.



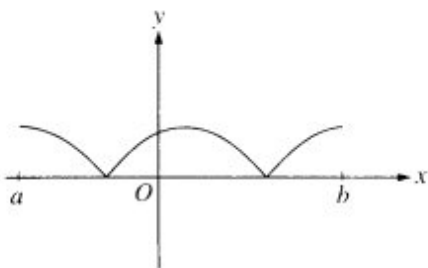
The graph of f is shown in the figure above. Which of the following could be the graph of derivative of f ?

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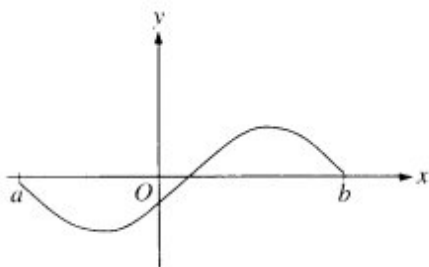
(A)



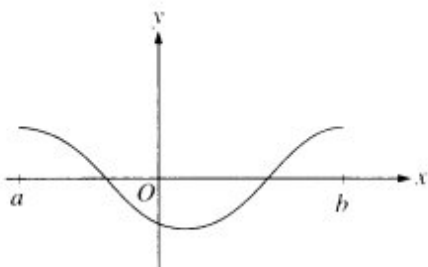
(B)



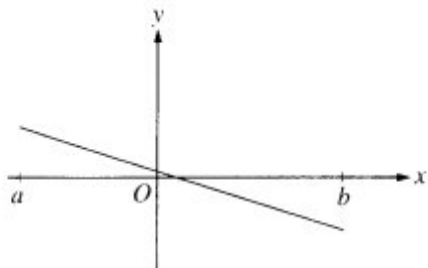
(C)



(D)

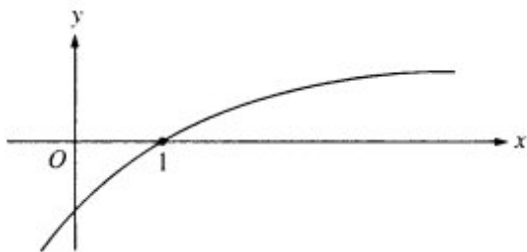


(E)



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15.

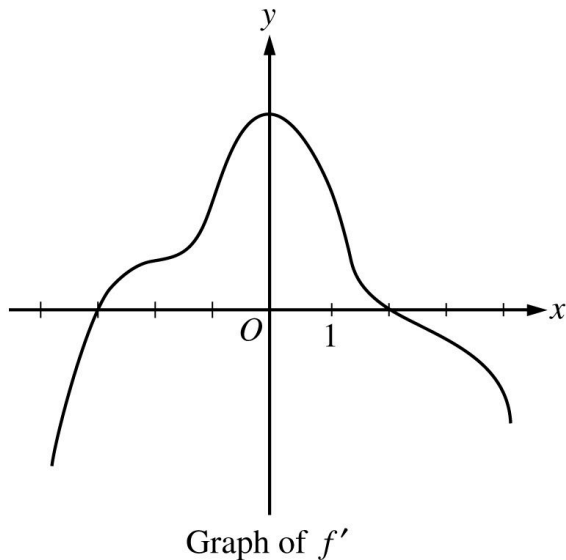


The graph of a twice-differentiable function f is shown in the figure above. Which of the following is true?

- (A) $f(1) < f'(1) < f''(1)$
- (B) $f(1) < f''(1) < f'(1)$
- (C) $f'(1) < f(1) < f''(1)$
- (D) $f''(1) < f(1) < f'(1)$
- (E) $f''(1) < f'(1) < f(1)$



16.



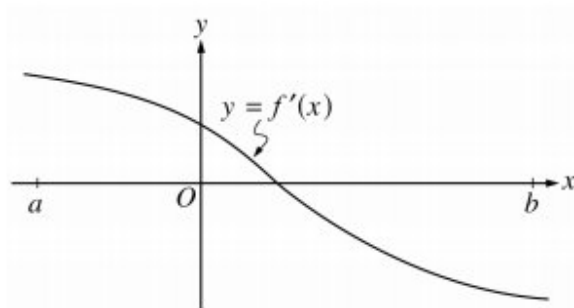
The graph of f' , the derivative of the function f , is shown above. Which of the following statements must be true?

- I. f has a relative minimum at $x = -3$.
 - II. The graph of f has a point of inflection at $x = -2$.
 - III. The graph of f is concave down for $0 < x < 4$.
- (A) I only
 - (B) II only
 - (C) III only
 - (D) I and II only
 - (E) I and III only



5-9-11

17.



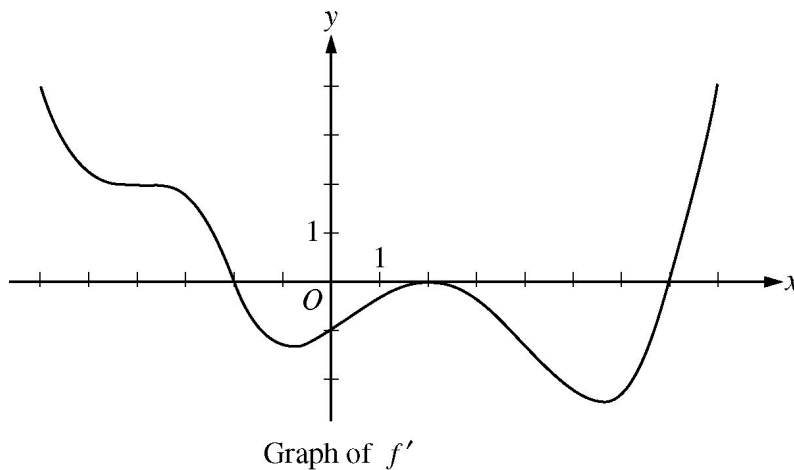
The graph of f' , the derivative of the function f , is shown in the figure above. Which of the following statements must be true?

- I. f is continuous on the open interval (a, b) .
- II. f is decreasing on the open interval (a, b) .
- III. The graph of f is concave down on the open interval (a, b) .

- (A) I only
- (B) I and II only
- (C) I and III only
- (D) II and III only



18.



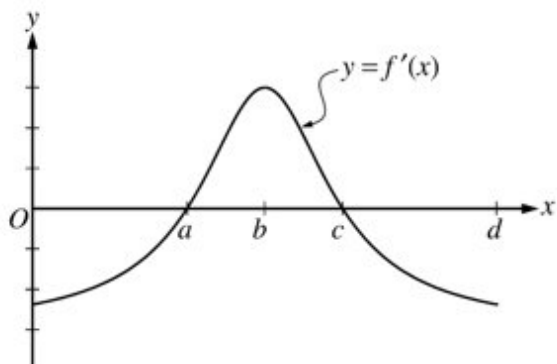
The figure above shows the graph of f' , the derivative of function f , for $-6 < x < 8$. Of the following, which best describes the graph of f on the same interval?

- (A) 1 relative minimum, 1 relative maximum, and 3 points of inflection
- (B) 1 relative minimum, 1 relative maximum, and 4 points of inflection
- (C) 2 relative minima, 1 relative maximum, and 2 points of inflection
- (D) 2 relative minima, 1 relative maximum, and 4 points of inflection
- (E) 2 relative minima, 2 relative maxima, and 3 points of inflection



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19.



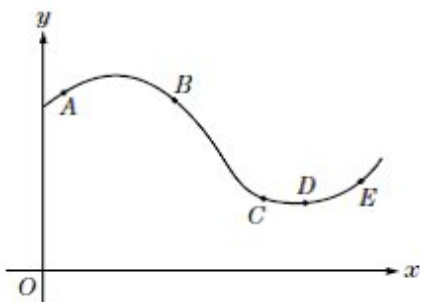
The graph of f' , the derivative of a function f , is shown above. The domain of f is the open interval $0 < x < d$. Which of the following statements is true?

- (A) f has a local minimum at $x = c$.
- (B) f has a local maximum at $x = b$.
- (C) The graph of f has a point of inflection at $(a, f(a))$.
- (D) The graph of f has a point of inflection at $(b, f(b))$.
- (E) The graph of f is concave up on the open interval (c, d) .



20.

At which of the five points on the graph in the figure at the right are $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ both negative?



- (A) A
- (B) B
- (C) C
- (D) D
- (E) E



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21.  Let g be the function given by $g(t) = 100 + 20 \sin\left(\frac{\pi t}{2}\right) + 10 \cos\left(\frac{\pi t}{6}\right)$ for $0 \leq t \leq 8$ is decreasing most rapidly when $t =$

(A) 0.949

(B) 2.017

(C) 3.106

(D) 5.965

(E) 8.000



22.

x	-2	0	3	5	6
$f'(x)$	3	1	4	7	5

Let f be a polynomial function with values of $f'(x)$ at selected values of x given in the table above. Which of the following must be true for $-2 < x < 6$?

(A) The graph of f is concave up.(B) The graph of f has at least two points of inflection.(C) f is increasing.(D) f has no critical points.(E) f has at least two relative extrema.

23. Let f be the function defined by $f(x) = 2x^3 - 3x^2 - 12x + 18$. On which of the following intervals is the graph of f both decreasing and concave up?

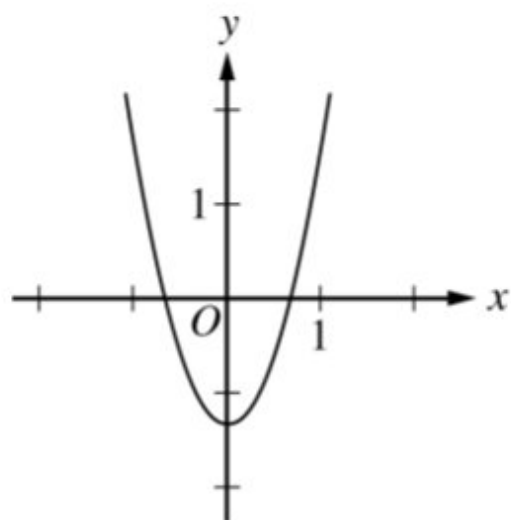
(A) $(-\infty, -1)$ (B) $(-1, \frac{1}{2})$ (C) $(-1, 2)$ (D) $(\frac{1}{2}, 2)$ (E) $(2, \infty)$ 

24. In the xy -plane, the graph of the twice-differentiable function $y = f(x)$ is concave up on the open interval $(0, 2)$ and is tangent to the line $y = 3x - 2$ at $x = 1$. Which of the following statements must be true about the derivative of f ?

(A) $f'(x) \leq 3$ on the interval $(0.9, 1)$.(B) $f'(x) \geq 3$ on the interval $(0.9, 1)$.(C) $f'(x) < 0$ on the interval $(0.9, 1.1)$.(D) $f'(x) > 0$ on the interval $(0.9, 1.1)$.(E) $f'(x)$ is constant on the interval $(0.9, 1.1)$.

5-9-11

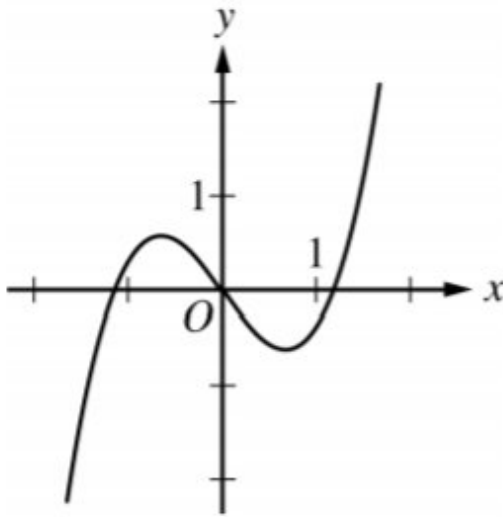
25.

Graph of f''

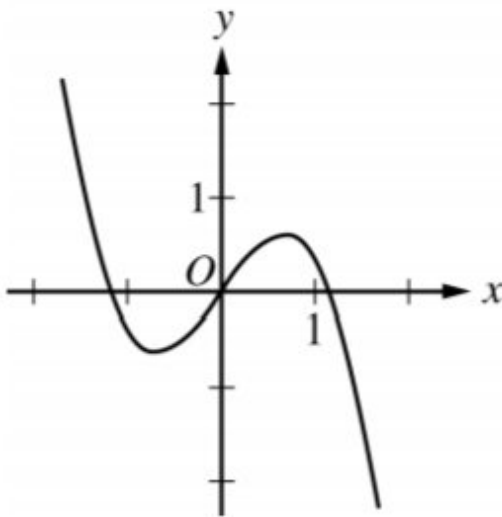
The graph of f'' , the second derivative of the function f , is shown above.
Which of the following could be the graph of f ?

5-9-11

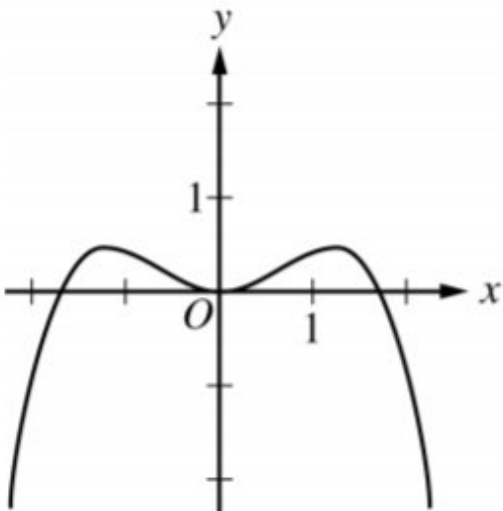
(A)



(B)

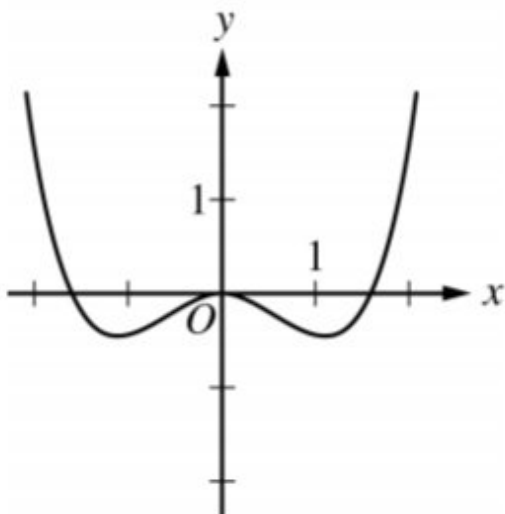


(C)

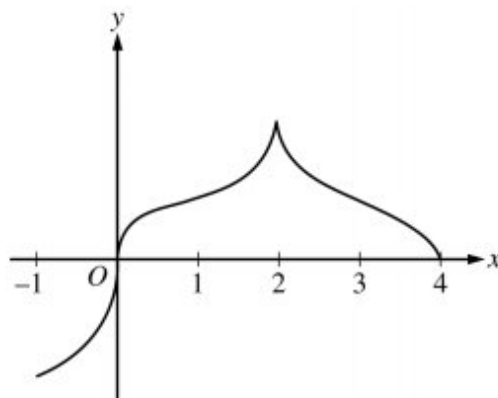


5-9-11

(D)



26.

Graph of f'

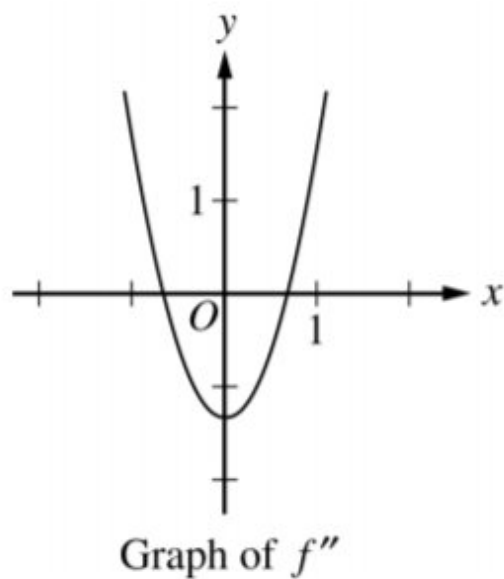
The graph of f' , the derivative of f , is shown above. The line tangent to the graph of f' at $x = 0$ is vertical, and f' is not differentiable at $x = 2$. Which of the following statements is true?

- (A) f' does not exist at $x = 2$
- (B) f is decreasing on the interval $(2, 4)$.
- (C) The graph of f has a point of inflection at $x = 2$.
- (D) The graph of f has a point of inflection at $x = 0$.
- (E) f has a local maximum at $x = 0$.



5-9-11

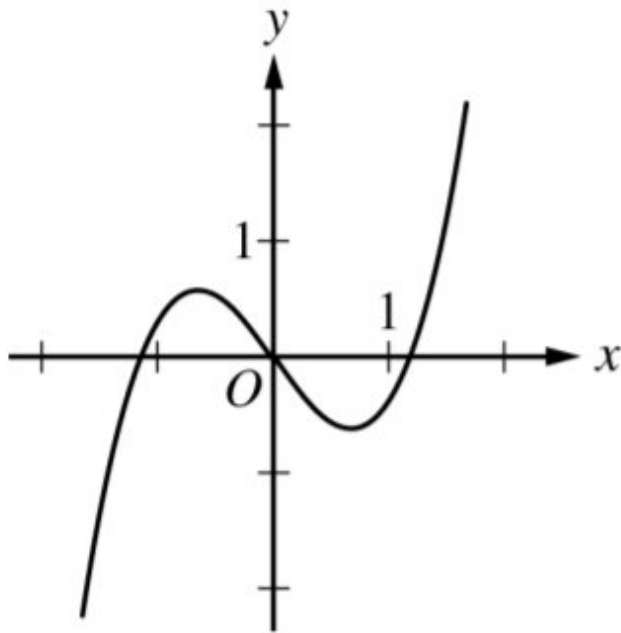
27.



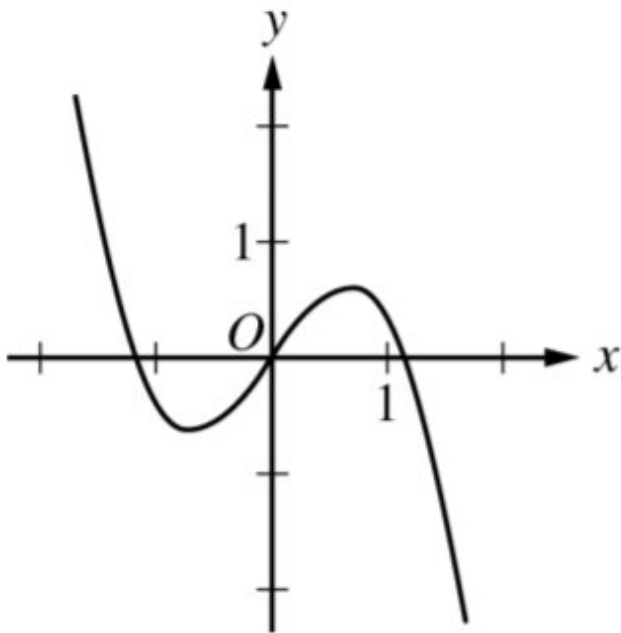
The graph of f'' , the second derivative of the function f , is shown above.
Which of the following could be the graph of f ?

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(A)

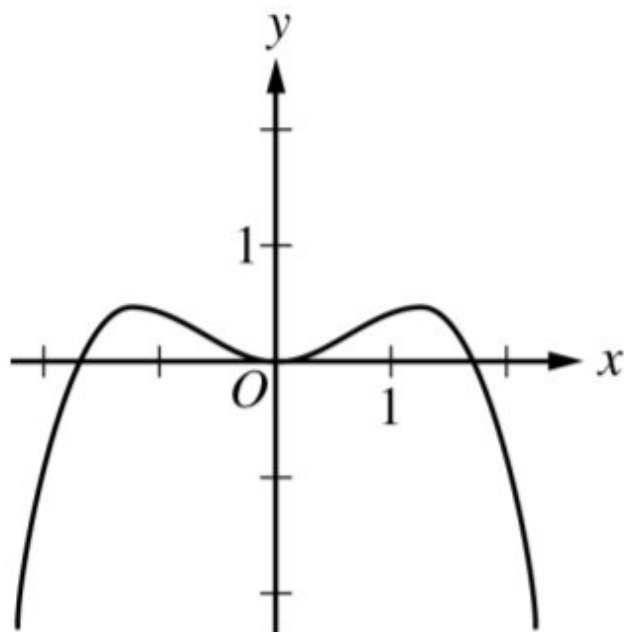


(B)

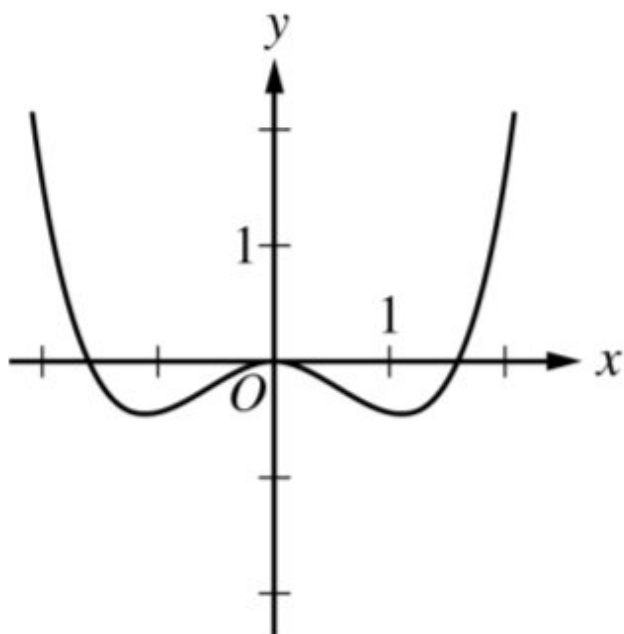


5-9-11

(C)

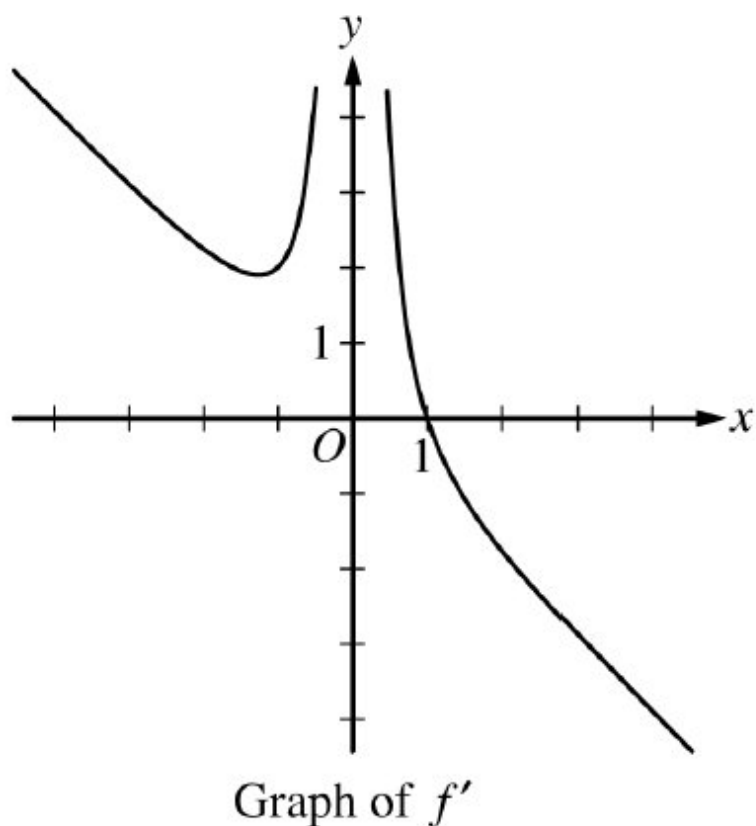


(D)



5-9-11

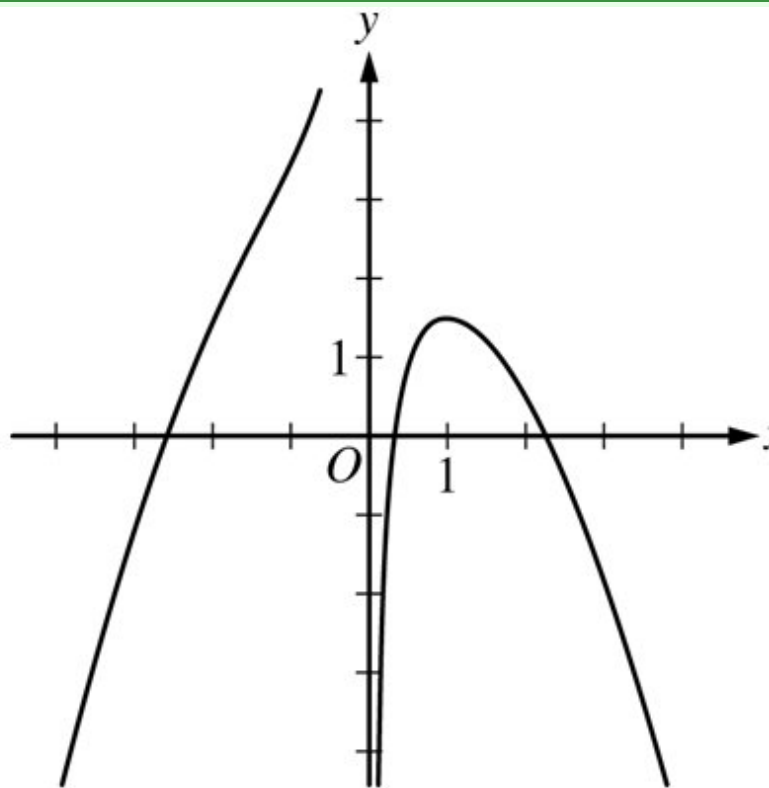
28.



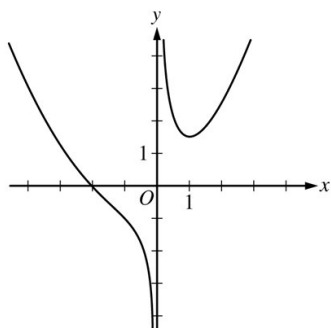
The graph of f' , the derivative of the function f , is shown above. Which of the following could be the graph of f ?

5-9-11

(A)

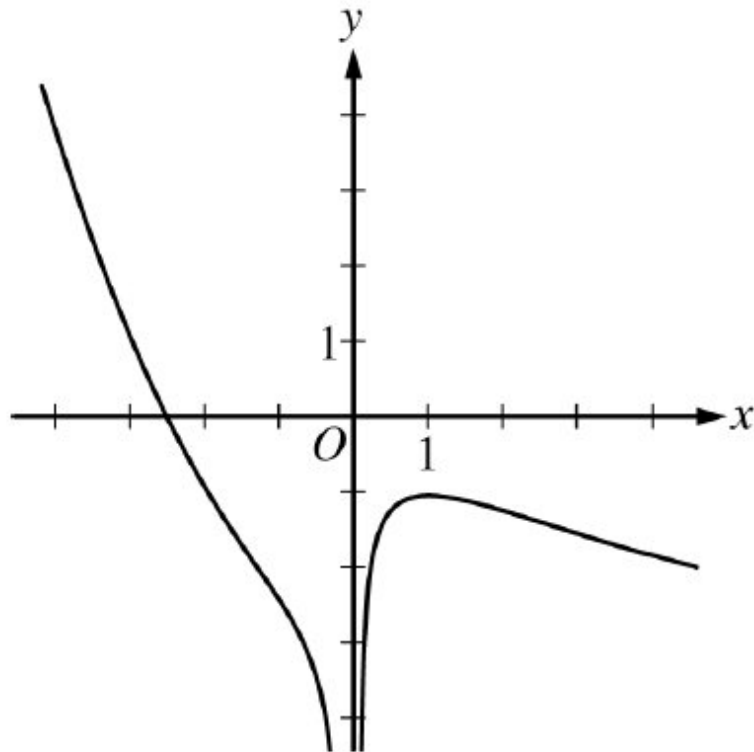


(B)

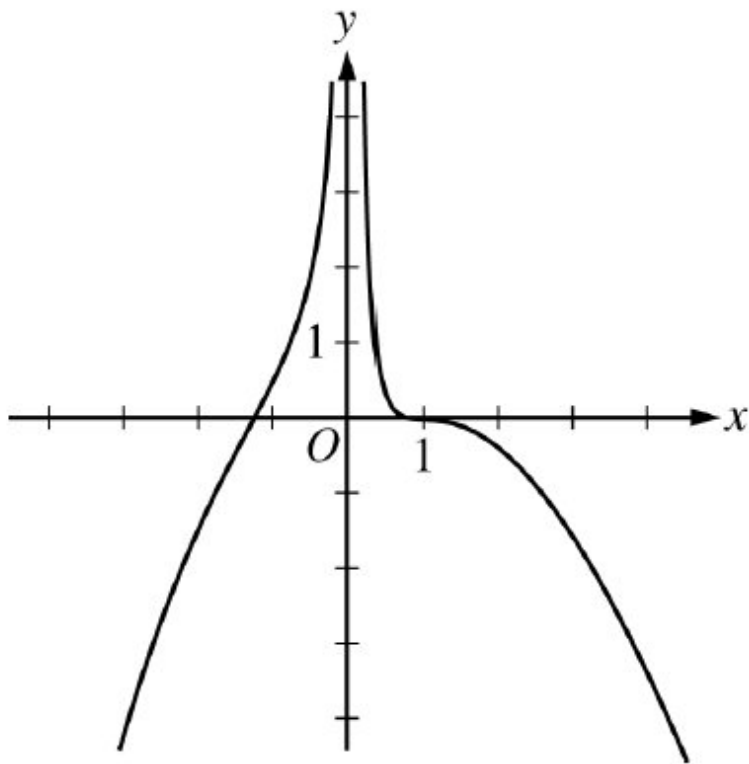


5-9-11

(C)



(D)



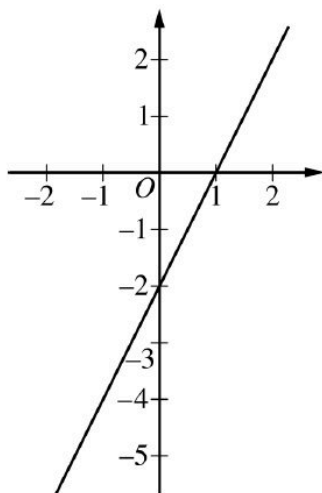
5-9-11

Answer A

This option is correct. Where the derivative f' is positive, the function f should be increasing. Where the derivative f' is negative, the function f should be decreasing. The derivative to the left of $x = 0$ is always positive. The derivative to the right of $x = 0$ is positive until $x = 1$, then negative. Therefore the graph of f should be increasing for $x < 0$ and $0 < x < 1$, and should be decreasing for $x > 1$. The graph in this option could therefore be the graph of f .

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29.

Graph of f

The graph of the function f is shown above for $-2 < x < 2$. Let g be the function defined by

$$g(x) = \int_0^x f(t) dt.$$

On what open interval is g negative and decreasing?

- (A) $-2 < x < 0$ only
(B) $-2 < x < 1$
(C) $0 < x < 1$ only
(D) $0 < x < 2$



Answer C

This option is correct. The function g is decreasing on the interval

$-2 < x < 1$ because $g' = f$ is negative on this interval. Because

$g(0) = 0$ and g is decreasing for $-2 < x < 1$, $g(x)$ must be positive for $-2 < x < 0$ and negative for $0 < x < 1$. On the interval $1 < x < 2$, the function g is increasing because $g' = f$ is positive on this interval. Therefore, g is negative and decreasing only for $0 < x < 1$.

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30.

x	10	11	12	13	14
$f(x)$	5	2	3	6	5

The table above gives values of the continuous function f at selected values of x .

If f has exactly two critical points on the open interval $(10, 14)$, which of the following must be true?

- (A) $f(x) > 0$ for all x in the open interval $(10, 14)$
- (B) $f'(x)$ exists for all x in the open interval $(10, 14)$
- (C) $f'(x) < 0$ for all x in the open interval $(10, 11)$
- (D) $f'(12) \neq 0$



5-9-11

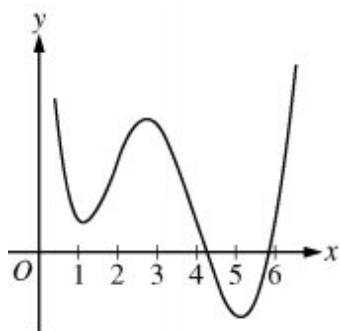
31.

x	1	2	3	4	5	6
$f(x)$	1	3	4	1	-2	1

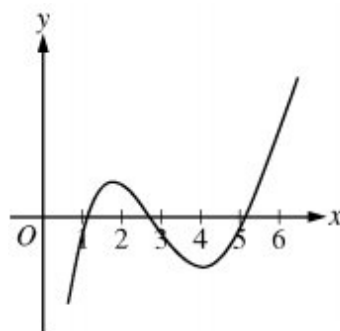
The function f is twice differentiable. Selected values of f are given in the table above. Which of the following could be the graph of $f''(x)$, the second derivative of f ?

5-9-11

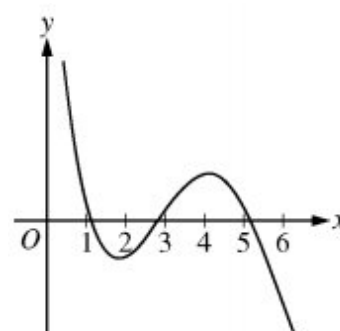
(A)



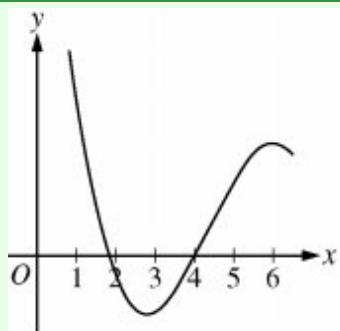
(B)



(C)

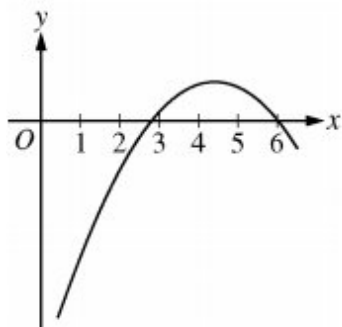


(D)

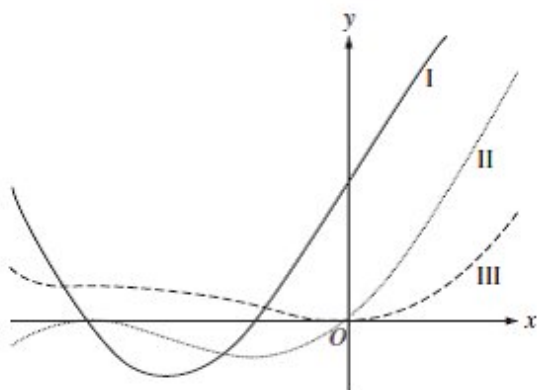


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(E)



32.



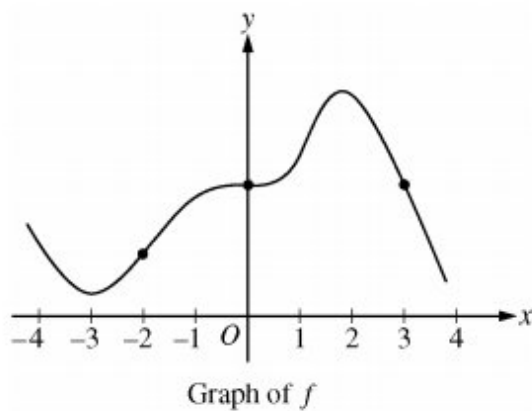
Three graphs labeled I, II, and III are shown above. One is the graph of f , one is the graph of f' , and one is the graph of f'' . Which of the following correctly identifies each of the three graphs?

(A)	f	f'	f''
	I	II	III
(B)	f	f'	f''
	I	III	II
(C)	f	f'	f''
	II	I	III
(D)	f	f'	f''
	II	III	I

(E)	f	f'	f''	✓
	III	II	I	

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33.



The graph of a differentiable function f is shown in the figure above.

Which of the following is true?

(A) $f'(-2) < f'(0) < f'(3)$

(B) $f'(-2) < f'(3) < f'(0)$

(C) $f'(3) < f'(-2) < f'(0)$

(D) $f'(3) < f'(0) < f'(-2)$

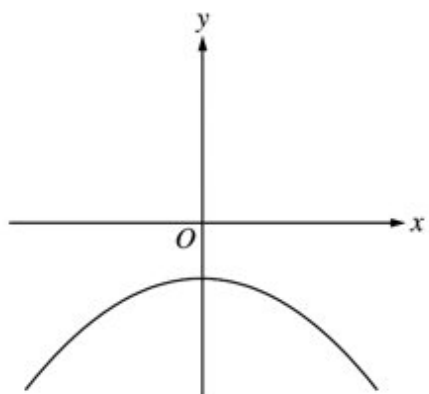


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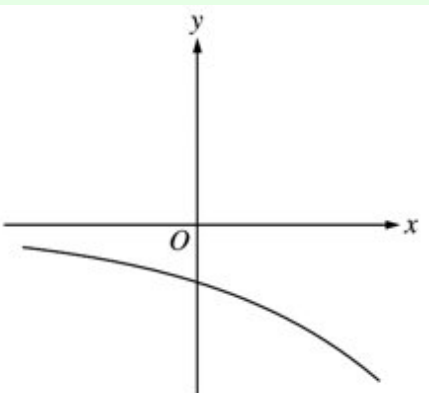
34. The function f has the property that $f(x)$, $f'(x)$ and $f''(x)$ are negative for all real values x . Which of the following could be the graph of f ?

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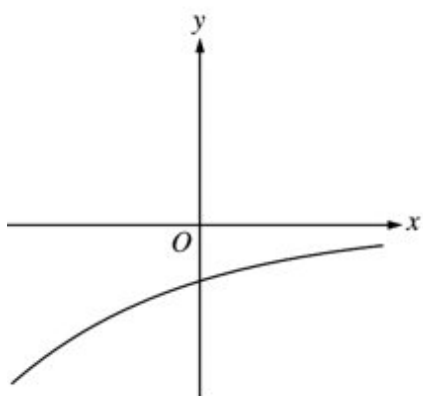
(A)



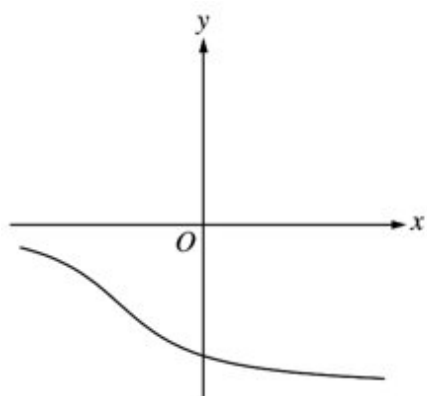
(B)



(C)

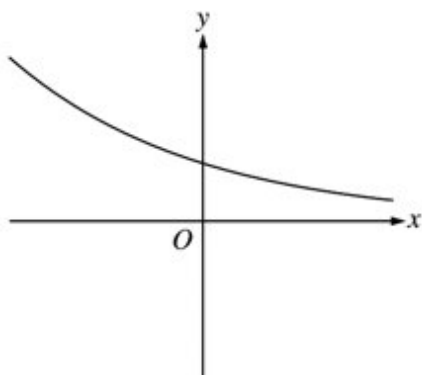


(D)

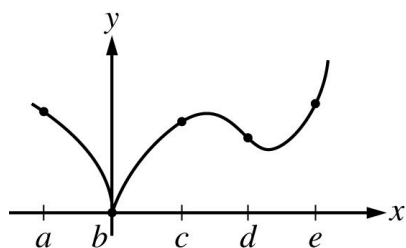


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(E)



35.

Graph of f

The graph of the function f is shown in the figure above. For which of the following values of x is $f'(x)$ positive and increasing?

- (A) a
- (B) b
- (C) c
- (D) d
- (E) e

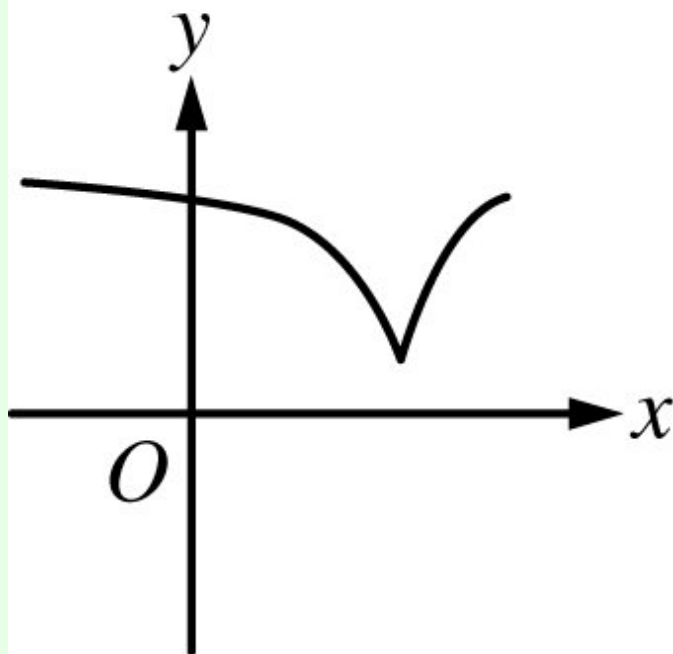


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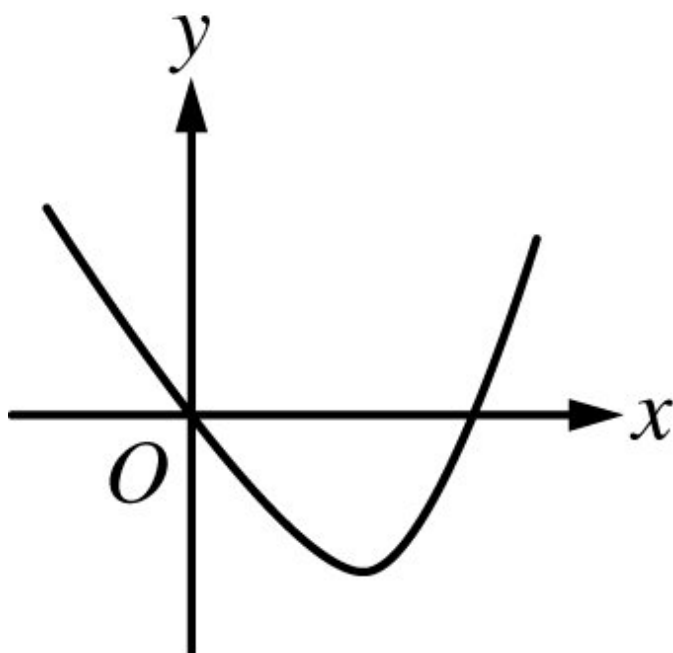
36. The function f is differentiable and increasing for all real numbers x , and the graph of f has exactly one point of inflection. Of the following, which could be the graph of f' , the derivative of f ?

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(A)

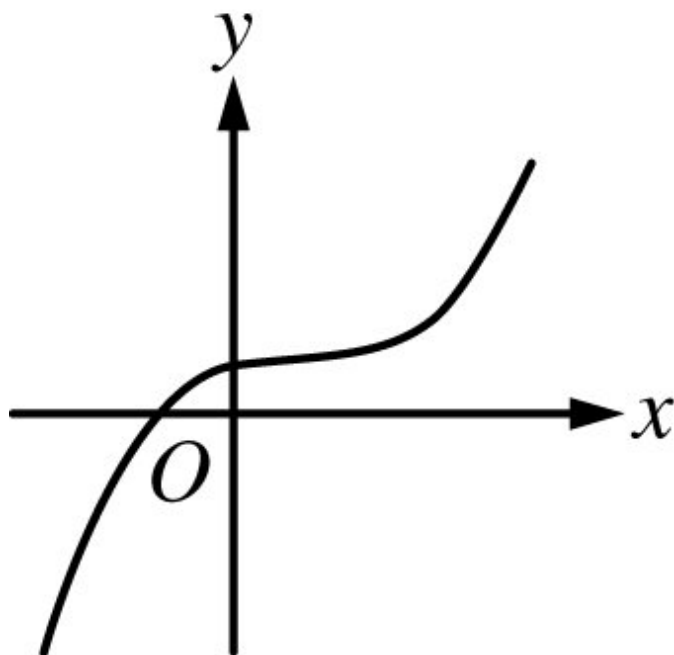


(B)

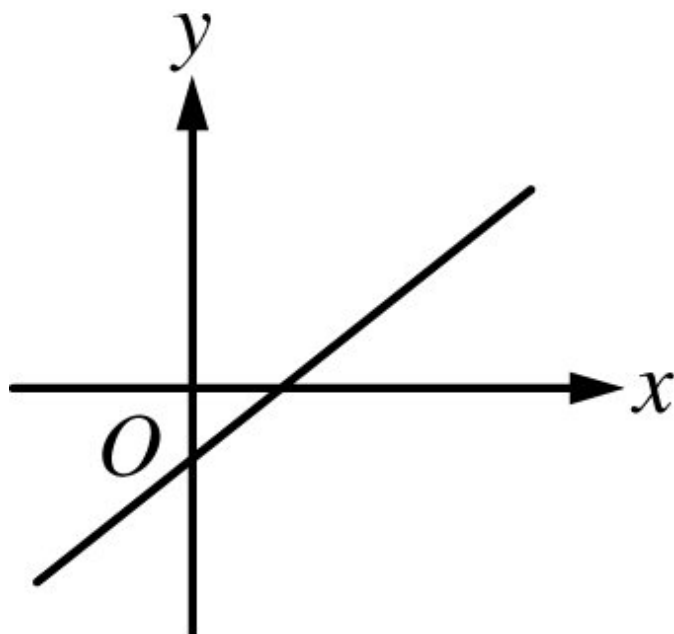


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(C)

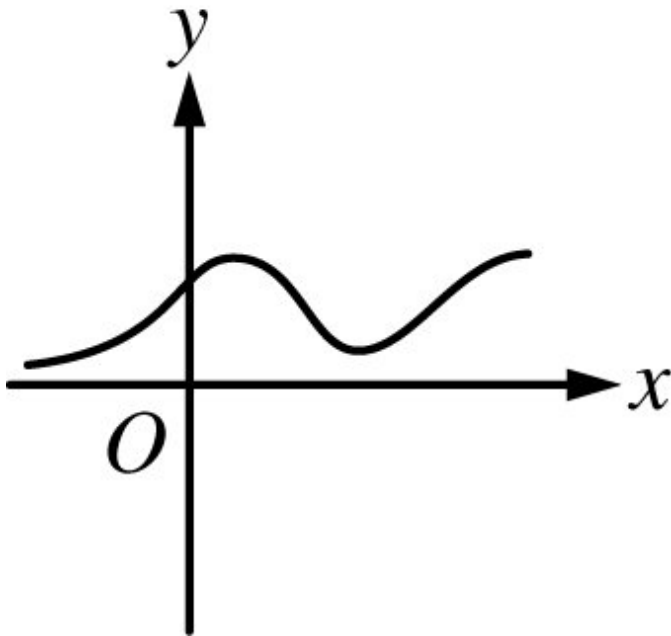


(D)



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(E)



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37. If $f'(x) > 0$ for all x and $f''(x) < 0$ for all x , which of the following could be a table of values for f ?

(A)

x	$f(x)$
-1	4
0	3
1	1

(B)

x	$f(x)$
-1	4
0	4
1	4

(C)

x	$f(x)$
-1	4
0	5
1	6

(D)

x	$f(x)$
-1	4
0	5
1	7

(E)

x	$f(x)$
-1	4
0	6
1	7



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38. Oil is leaking from a pipeline on the surface of a lake and forms an oil slick whose volume increases at a constant rate of 2000 cubic centimeters per minute. The oil slick takes the form of a right circular cylinder with both its radius and height changing with time. (Note: The volume V of a right circular cylinder with radius r and height h is given by $V = \pi r^2 h$.)

A recovery device arrives on the scene and begins removing oil. The rate at which is removed is $R(t) = 400\sqrt{t}$ cubic centimeters per minute, where t is the time in minutes since the device began working. Oil continues to leak at the rate of 2000 cubic centimeters per minute. Find the time t when the oil slick reaches its maximum volume. Justify your answer.

Part B

The response can earn up to 3 points:

1 point: For $R(t) = 2000$

$$\frac{dV}{dt} = 2000 - R(t), \text{ so } \frac{dV}{dt} = 0 \text{ when } R(t) = 2000.$$

1 point: For correct answer

This occurs when $t = 25$ minutes.

1 point: For the justification

Since $\frac{dV}{dt} > 0$ for $0 < t < 25$ and $\frac{dV}{dt} < 0$ for $t > 25$, the oil slick reaches its maximum volume 25 minutes after the device begins working.



0	1	2	3
---	---	---	---

The response can earn up to 3 points:

1 point: For $R(t) = 2000$

$$\frac{dV}{dt} = 2000 - R(t), \text{ so } \frac{dV}{dt} = 0 \text{ when } R(t) = 2000.$$

1 point: For correct answer

This occurs when $t = 25$ minutes.

1 point: For the justification

Since $\frac{dV}{dt} > 0$ for $0 < t < 25$ and $\frac{dV}{dt} < 0$ for $t > 25$, the oil slick reaches its maximum volume 25 minutes after the device begins working.

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39.  The derivative of the function f is given by $f'(x) = x^2 \cos(x^2)$. How many points of inflection does the graph of f have on the open interval $(-2, 2)$?

(A) One
(B) Two
(C) Three
(D) Four

(E) Five



40. Which of the following statements about the function given by $f(x) = x^4 - 2x^3$ is true?

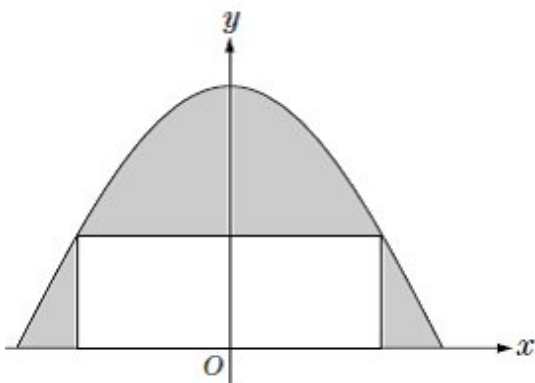
(A) The function has no relative extremum.
(B) The graph of the function has one point of inflection and the function has two relative extrema.

(C) The graph of the function has two points of inflection and the function has one relative extremum.



(D) The graph of the function has two points of inflection and the function has two relative extrema.
(E) The graph of the function has two points of inflection and the function has three relative extrema.

41.



A rectangle with one side on the x -axis has its upper vertices on the graph of $y = \cos x$, as shown in the figure above. What is the minimum area of the shaded region?

(A) 0.799

(B) 0.878



(C) 1.140

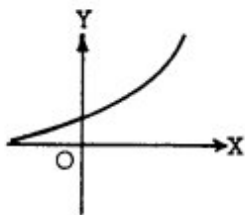
(D) 1.439

(E) 2.000

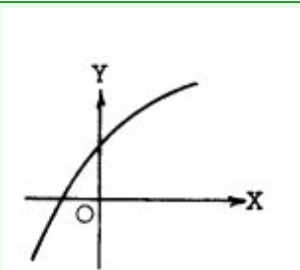
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42. If y is a function of x such that $y' > 0$ for all x and $y'' < 0$ for all x , which of the following could be part of the graph of $y = f(x)$?

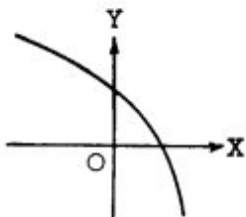
(A)



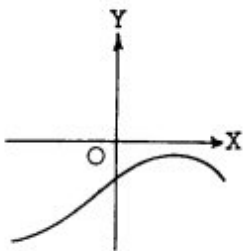
(B)



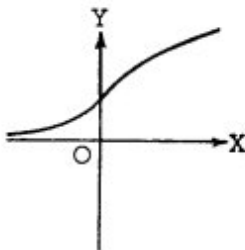
(C)



(D)

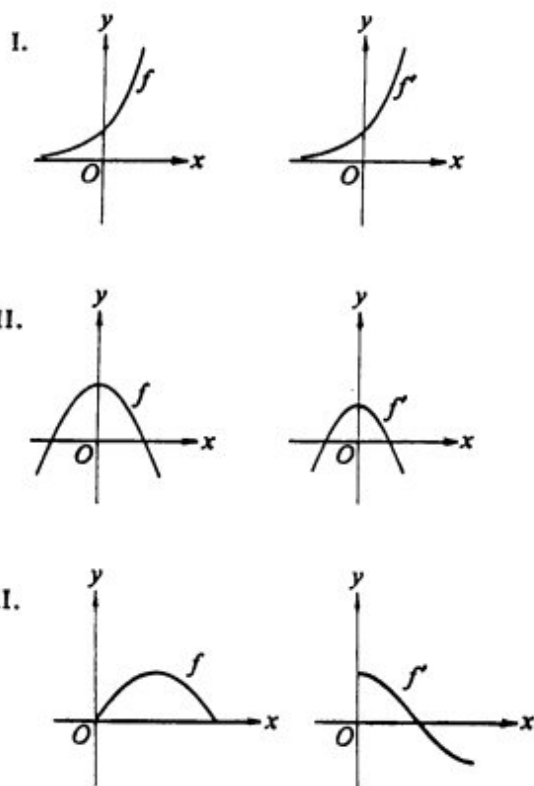


(E)



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43.



Which of the following pairs of graphs could represent the graph of a function and the graph of its derivative?

- (A) I only
(B) II only
(C) III only
(D) II and III
(E) I and III



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44. Let F be a function defined for all real numbers x such that $F'(x) > 0$ and $F''(x) > 0$. Which of the following could be a table of values for F ?

(A)

x	$F(x)$
1	-3
2	-4
3	-6
4	-9

(B)

x	$F(x)$
1	-3
2	-1
3	3
4	19



(C)

x	$F(x)$
1	-3
2	0
3	3
4	6

(D)

x	$F(x)$
1	-3
2	5
3	11
4	13

(E)

x	$F(x)$
1	-3
2	-4
3	-3
4	0

45. The volume of a cylindrical tin can with a top and a bottom is to be 16π cubic inches. If a minimum amount of tin is to be used to construct the can, what must be the height, in inches, of the can?

(A) $2\sqrt[3]{2}$

(B) $2\sqrt{2}$

(C) $2\sqrt[3]{4}$

(D) 4



(E) 8

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46. The speed of a runner, in miles per hour, on a straight trail is modeled by $f(m) = \frac{1}{10}(-2m^3 + 9m^2 - 12m) + 7$, where m is the runner's distance, in miles, from the start of the trail. What is the maximum speed of the runner for $0 \leq m \leq 3$?
- (A) 6.5
(B) 6.6
(C) 7.0
(D) 7.5

Answer C

This option is correct. Because f is continuous, the Extreme Value Theorem guarantees the existence of a maximum speed on the closed interval $0 \leq m \leq 3$, and that absolute maximum will occur at a critical value or at one of the endpoints.

$$f'(m) = \frac{1}{10}(-6m^2 + 18m - 12) = -\frac{6}{10}(m^2 - 3m + 2) = -\frac{6}{10}(m - 1)(m - 2)$$

$$(m - 1)(m - 2) = 0$$

The critical values are $m = 1$ and $m = 2$. Therefore the candidates are $m = 0, 1, 2$, and 3 . Evaluate f at each candidate and select the largest.

$$f(0) = 7.0$$

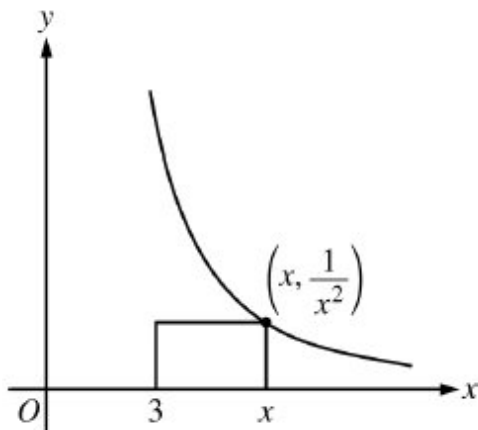
$$f(1) = 6.5$$

$$f(2) = 6.6$$

$$f(3) = 6.1$$

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47.



The points $(3, 0)$, $(x, 0)$, $(x, 1/x^2)$, and $(3, 1/x^2)$ are the vertices of a rectangle, where $x \geq 3$, as shown in the figure above. For what value of x does the rectangle have a maximum area?

- (A) 3
- (B) 4
- (C) 6
- (D) 9
- (E) There is no such value of x .



48. A rectangular area is to be enclosed by a wall on one side and fencing on the other three sides. If 18 meters of fencing are used, what is the maximum area that can be enclosed?

- (A) $\frac{9}{2} m^2$
- (B) $\frac{81}{4} m^2$
- (C) $27 m^2$
- (D) $40 m^2$
- (E) $\frac{81}{2} m^2$



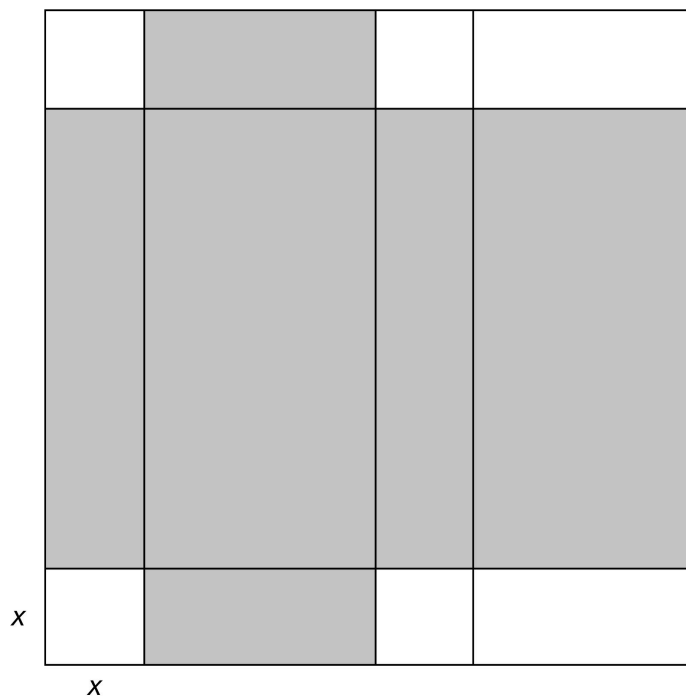
49.  Consider all right circular cylinders for which the sum of the height and circumference is 30 centimeters. What is the radius of the one with maximum volume?

- (A) 3 cm
- (B) 10 cm
- (C) 20 cm
- (D) $\frac{30}{\pi^2}$ cm
- (E) $\frac{10}{\pi}$ cm



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50.



The figure above represents a square sheet of cardboard with side length 40 inches. The sheet is cut and pieces are discarded. When the cardboard is folded, it becomes a rectangular box with a lid. The pattern for the rectangular box with a lid is shaded in the figure. Four squares with side length x and two rectangular regions are discarded from the cardboard. Which of the following statements is true? (The volume V of a rectangular box is given by $V = lwh$.)

- (A) When $x = 20$ inches, the box has a minimum possible volume.
- (B) When $x = 20$ inches, the box has a maximum possible volume.
- (C) When $x = \frac{20}{3}$ inches, the box has a minimum possible volume.
- (D) When $x = \frac{20}{3}$ inches, the box has a maximum possible volume. ✓

Answer D

Correct. To find the volume of the rectangular box, the length, width, and height of the box must be identified. When the shaded region is folded to form the box, the height of the box is x inches. Since the side length of the cardboard sheet is 40 inches, the length of the box is $(40 - 2x)$ inches because each of the smaller squares has side length x . Label the width of the rectangular box as y inches. The side length of the cardboard sheet, in inches, can also be defined as $x + y + x + y = 2x + 2y = 40$, so $x + y = 20 \Rightarrow y = 20 - x$ is an expression for the width of the box in inches. Therefore, the volume of the box, in cubic inches, is $x(40 - 2x)(20 - x)$. To find the value of x that maximizes the volume, take the derivative of the expression. Since this is a polynomial, it might be easier to expand the polynomial before taking the derivative. This results in $2x^3 - 80x^2 + 800x$. Using the power rule, the derivative is $6x^2 - 160x + 800$. Factoring this quadratic polynomial as $2(3x^2 - 80x + 400) = 2(3x - 20)(x - 20) \Rightarrow 2(3x - 20)(x - 20) = 0 \Rightarrow x = \frac{20}{3}$ produces the maximum volume of the box.

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51. The acceleration, in meters per second per second, of a race car is modeled by $A(t) = t^3 - \frac{15}{2}t^2 + 12t + 10$, where t is measured in seconds. What is the car's maximum acceleration on the time interval $0 \leq t \leq 6$?
- (A) The maximum acceleration of the race car is 2 meters per second per second and occurs at $t = 4$ seconds.
- (B) The maximum acceleration of the race car is 6 meters per second per second and occurs at $t = 28$ seconds.
- (C) The maximum acceleration of the race car is 15.5 meters per second per second and occurs at $t = 1$ second.
- (D) The maximum acceleration of the race car is 28 meters per second per second and occurs at $t = 6$ seconds. ✓

Answer D

Correct. The Extreme Value Theorem guarantees that there will be a maximum value of $A(t)$ on the closed interval $[0, 6]$. This will occur at a critical point or an endpoint.

$$A'(t) = 3t^2 - 15t + 12 = 3(t - 1)(t - 4)$$

The critical values are $t = 1$ and $t = 4$. Therefore, the candidates are $t = 0$, $t = 1$, $t = 4$, and $t = 6$. $A(t)$ is evaluated at each candidate.

$$A(0) = 10$$

$$A(1) = 15.5$$

$$A(4) = 2$$

$$A(6) = 28$$

The maximum acceleration is 28 meters per second per second.

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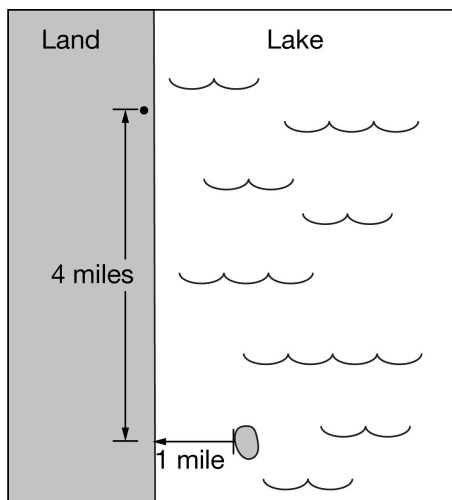
52. The total cost, in dollars, to order x units of a certain product is modeled by $C(x) = 5x^2 + 320$. According to the model, for what size order is the cost per unit a minimum?
- (A) An order of 1 unit has a minimum cost per unit.
- (B) An order of 8 units has a minimum cost per unit. ✓
- (C) An order of 80 units has a minimum cost per unit.
- (D) An order of 320 units has a minimum cost per unit.

Answer B

Correct. If the total cost to order x units is given by $C(x) = 5x^2 + 320$, the cost per unit is given by $\frac{5x^2+320}{x} = 5x + \frac{320}{x}$. Based on the context, a reasonable domain of this expression is $x > 0$. The minimum cost per unit is thus the minimum value of this expression. It follows that $\frac{d}{dx} \left(5x + \frac{320}{x} \right) = 5 - \frac{320}{x^2}$, which equals 0 when $x = \pm 8$. Since only the critical point $x = 8$ is in the domain, it is the only candidate for the location of the minimum value. The cost per unit for an order of 8 units is $\left(5x + \frac{320}{x} \right) \Big|_{x=8} = 80$ dollars. An order of 8 units has a minimum cost per unit.

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53.



An electrical power station is located on the edge of a lake, as shown in the figure above. An electrical line needs to be connected from the station to an island in the lake that is located 4 miles due south and 1 mile due east of the station. It costs 5,000 per mile to install an electrical line on land and 10,000 per mile to install an electrical line underwater. If C represents a cost function, which of the following methods best explains how to determine the minimum cost, in dollars, for connecting the electrical line from the station to the island?

- (A) Let $C(x) = 10,000\sqrt{(4-x)^2 + 1} + 5,000x$. Solve $C'(x) = 0$ and find the values of x where $C'(x)$ changes sign from positive to negative. Evaluate C for those values of x to determine the minimum cost.
- (B) Let $C(x) = 10,000\sqrt{(4-x)^2 + 1} + 5,000x$. Solve $C'(x) = 0$ and find the values of x where $C'(x)$ changes sign from negative to positive. Evaluate C for those values of x to determine the minimum cost. ✓
- (C) Let $C(x) = 10,000\sqrt{x^2 + 1} + 5,000x$. Solve $C'(x) = 0$ and find the values of x where $C'(x)$ changes sign from positive to negative. Evaluate C for those values of x to determine the minimum cost.
- (D) Let $C(x) = 10,000\sqrt{x^2 + 1} + 5,000x$. Solve $C'(x) = 0$ and find the values of x where $C'(x)$ changes sign from negative to positive. Evaluate C for those values of x to determine the minimum cost.

Answer B

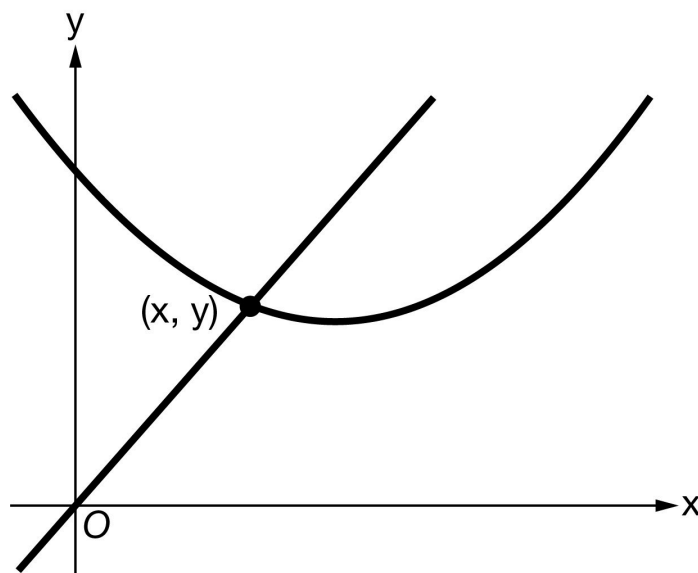
Correct. First, express the cost of the electrical line as a function by considering the distance, in miles, of each section of the electrical line (underwater and on land) and the cost per mile for installation. Consider a right triangle with legs of length 1 and $(4 - x)$, where x represents the land distance, in miles, for the electrical line. The top vertex of the triangle is the point at which the land section of the electrical line connects to the underwater section of the line. The underwater electrical line distance, in miles, is the hypotenuse of the right triangle $\sqrt{(4 - x)^2 + 1}$. The cost of the electrical line underwater can be expressed by $10,000\sqrt{(4 - x)^2 + 1}$. The cost of the electrical line on land can be expressed by $5,000x$. The total cost is $C(x) = 10,000\sqrt{(4 - x)^2 + 1} + 5,000x$. To determine the minimum cost,

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take the derivative of C and solve $C'(x) = 0$. Find all x values where there is a relative minimum, where $C'(x)$ changes sign from negative to positive. Evaluate C for those values of x to determine the minimum cost.

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54.



Consider all lines in the xy -plane that pass through both the origin and a point (x, y) on the graph of $y = x^2 - 4x + 9$ for $1 \leq x \leq 4$. The figure above shows one such line and the graph of $y = x^2 - 4x + 9$. Which of the following statements is true?

- (A) The line with minimum slope passes through the graph of $y = x^2 - 4x + 9$ at $x = 1$.
- (B) The line with minimum slope passes through the graph of $y = x^2 - 4x + 9$ at $x = 2$.
- (C) The line with minimum slope passes through the graph of $y = x^2 - 4x + 9$ at $x = 3$. ✓
- (D) The line with minimum slope passes through the graph of $y = x^2 - 4x + 9$ at $x = 4$.

Answer C

Correct. The first step is to express the slope of a line through the origin and a point on the curve in terms of the variable x .

$$m = \frac{y}{x} = \frac{x^2 - 4x + 9}{x} = x - 4 + \frac{9}{x}$$

The Extreme Value Theorem guarantees that there will be a minimum value and a maximum value of m on the closed interval $1 \leq x \leq 4$ and that they will occur at a critical point or at an endpoint.

$$\frac{d}{dx} m = 1 - \frac{9}{x^2} = 0 \Rightarrow x = 3 \text{ is the only critical point in the interval } 1 \leq x \leq 4.$$

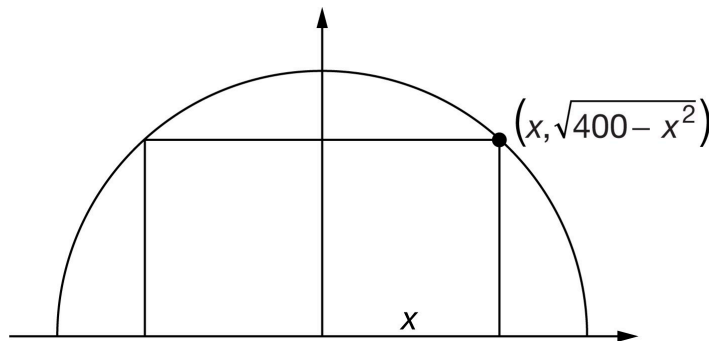
Evaluating m at the critical point and at each endpoint gives the following.

x	m
1	6
3	2
4	$\frac{9}{4}$

The minimum slope is $m = 2$, which occurs at $x = 3$.

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55.



The figure above shows a rectangle inscribed in a semicircle with a radius of 20. The area of such a rectangle is given by $A(x) = 2x\sqrt{400 - x^2}$, where the width of the rectangle is $2x$. It can be shown that $A'(x) = \frac{-2x^2}{\sqrt{400 - x^2}} + 2\sqrt{400 - x^2}$ and A has critical values of -20 , $-10\sqrt{2}$, $10\sqrt{2}$, and 20 . It can also be shown that $A'(x)$ changes from positive to negative at $x = 10\sqrt{2}$. Which of the following statements is true?

- (A) The inscribed rectangle with maximum area has dimensions $10\sqrt{2}$ by $10\sqrt{2}$.
- (B) The inscribed rectangle with minimum area has dimensions $10\sqrt{2}$ by $10\sqrt{2}$.
- (C) The inscribed rectangle with maximum area has dimensions $20\sqrt{2}$ by $10\sqrt{2}$. ✓
- (D) The inscribed rectangle with minimum area has dimensions $20\sqrt{2}$ by $10\sqrt{2}$.

Answer C

Correct. The feasible domain of A is the open interval $0 < x < 20$. Because $x = 10\sqrt{2}$ is the only critical value in this interval and $A'(x)$ changes from positive to negative at $x = 10\sqrt{2}$, A attains an absolute maximum value at this x -value. The corresponding y -value is $10\sqrt{2}$. Since the width of the rectangle is $2x$, the inscribed rectangle with maximum area has dimensions $20\sqrt{2}$ by $10\sqrt{2}$.

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56. The acceleration, in centimeters per second per second, of a projectile is modeled by $A(t) = 2t^3 - 15t^2 + 36t$, where t is measured in seconds. What is the projectile's maximum acceleration on the time interval $0 \leq t \leq 4$?
- (A) The maximum acceleration of the projectile is 4 centimeters per second per second and occurs at $t = 32$ seconds.
- (B) The maximum acceleration of the projectile is 27 centimeters per second per second and occurs at $t = 3$ seconds.
- (C) The maximum acceleration of the projectile is 28 centimeters per second per second and occurs at $t = 2$ seconds.
- (D) The maximum acceleration of the projectile is 32 centimeters per second per second and occurs at $t = 4$ seconds. ✓

Answer D

Correct. The Extreme Value Theorem guarantees that there will be a maximum value of $A(t)$ on the closed interval $[0, 4]$. This will occur at a critical point or an endpoint.

$$A'(t) = 6t^2 - 30t + 36 = 6(t^2 - 5t + 6) = 6(t - 2)(t - 3)$$

The critical values are $t = 2$ and $t = 3$. Therefore, the candidates are $t = 0$, $t = 2$, $t = 3$, and $t = 4$. $A(t)$ is evaluated at each candidate.

$$A(0) = 0$$

$$A(2) = 28$$

$$A(3) = 27$$

$$A(4) = 32$$

The maximum acceleration is 32 centimeters per second per second and occurs at $t = 4$ seconds.

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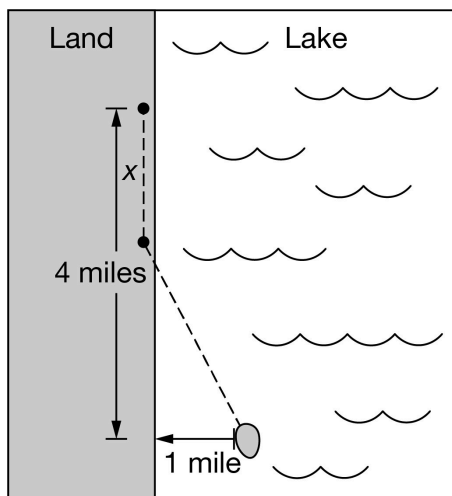
57. The total cost, in dollars, to order x units of a certain product is modeled by $C(x) = 7x^2 + 252$. According to the model, for what size order is the cost per unit a minimum?
- (A) An order of 1 unit has a minimum cost per unit.
- (B) An order of 6 units has a minimum cost per unit. ✓
- (C) An order of 84 units has a minimum cost per unit.
- (D) An order of 252 units has a minimum cost per unit.

Answer B

Correct. If the total cost to order x units is given by $C(x) = 7x^2 + 252$, the cost per unit is given by $\frac{7x^2+252}{x} = 7x + \frac{252}{x}$. Based on the context, a reasonable domain of this expression is $x > 0$. The minimum cost per unit is thus the minimum value of this expression. It follows that $\frac{d}{dx}\left(7x + \frac{252}{x}\right) = 7 - \frac{252}{x^2}$, which equals 0 when $x = \pm 6$. Since only the critical point $x = 6$ is in the domain, it is the only candidate for the location of the minimum value. An order of 6 units has a minimum cost per unit, and the cost per unit for an order of 6 units is $\left(7x + \frac{252}{x}\right)\bigg|_{x=6} = 84$ dollars.

5-9-11

58.



An athlete is planning for the “Land and Lake Race,” the path of which is shown in the figure above. The contestants will start at the dot, run up to 4 miles along the beach, enter the water at any time, and swim to the island. The athlete estimates that they can run along the beach at a constant rate of 8 miles per hour and swim at a constant rate of 2 miles per hour. Let T be a function that represents the time to complete the race, where x is the distance in miles that the athlete runs and $T(x)$ is measured in hours. Which of the following methods best explains how to determine the minimum time, in hours, the athlete should run along the beach and then swim to the island?

- (A) Let $T(x) = 8x + 2\sqrt{1^2 + (4-x)^2}$. Solve $T'(x) = 0$ and find the values of x where $T'(x)$ changes sign from negative to positive. Evaluate T for those values of x to determine the minimum time.
- (B) Let $T(x) = \frac{x}{8} + \frac{\sqrt{1^2 + (4-x)^2}}{2}$. Solve $T'(x) = 0$ and find the values of x where $T'(x)$ changes sign from negative to positive. Evaluate T for those values of x to determine the minimum time. ✓
- (C) Let $T(x) = \frac{x}{8} + \frac{\sqrt{1^2 + x^2}}{2}$. Solve $T'(x) = 0$ and find the values of x where $T'(x)$ changes sign from negative to positive. Evaluate T for those values of x to determine the minimum time.
- (D) Let $T(x) = \frac{x}{8} + \frac{\sqrt{(4-x)^2}}{2}$. Solve $T'(x) = 0$ and find the values of x where $T'(x)$ changes sign from negative to positive. Evaluate T for those values of x to determine the minimum time.

Answer B

Correct. First, express the total time of the race by considering the time run on the beach plus the time spent swimming. Consider a right triangle with legs of length 1 and $(4-x)$, where x represents the distance, in miles, run on the beach. The top vertex of the triangle represents the point where the athlete enters the water and begins swimming. The length of the hypotenuse is $\sqrt{1^2 + (4-x)^2}$. Use the relationship that time is distance divided by rate. The time spent

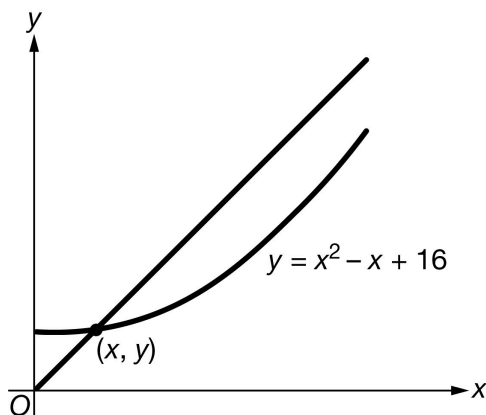
running on the beach is $\frac{x}{8}$ and the time spent swimming is $\frac{\sqrt{1^2 + (4-x)^2}}{2}$, and the total time is

$T(x) = \frac{x}{8} + \frac{\sqrt{1^2 + (4-x)^2}}{2}$. To determine the minimal time, take the derivative of T and solve $T'(x) = 0$. Find all x -

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values where there is a relative minimum, where $T'(x)$ changes from negative to positive. Evaluate T for those values of x to determine the minimal time.

59.



Consider all lines in the xy -plane that pass through both the origin and a point (x, y) on the graph of $y = x^2 - x + 16$ for $1 \leq x \leq 8$. The figure above shows one such line and the graph of $y = x^2 - x + 16$. Which of the following statements is true?

- (A) The line with minimum slope passes through the graph of $y = x^2 - x + 16$ at $x = 1$.
- (B) The line with minimum slope passes through the graph of $y = x^2 - x + 16$ at $x = 4$. ✓
- (C) The line with minimum slope passes through the graph of $y = x^2 - x + 16$ at $x = 7$.
- (D) The line with minimum slope passes through the graph of $y = x^2 - x + 16$ at $x = 8$.

Answer B

Correct. The first step is to express the slope of a line through the origin and a point on the curve in terms of the variable x .

$$m = \frac{y}{x} = \frac{x^2 - x + 16}{x} = x - 1 + \frac{16}{x}$$

The Extreme Value Theorem guarantees that there will be a minimum value and a maximum value of m on the closed interval $1 \leq x \leq 8$ and that they will occur at a critical point or at an endpoint.

$$\frac{dm}{dx} = 1 - \frac{16}{x^2} = 0 \Rightarrow x = 4 \text{ is the only critical point in the interval } 1 \leq x \leq 8.$$

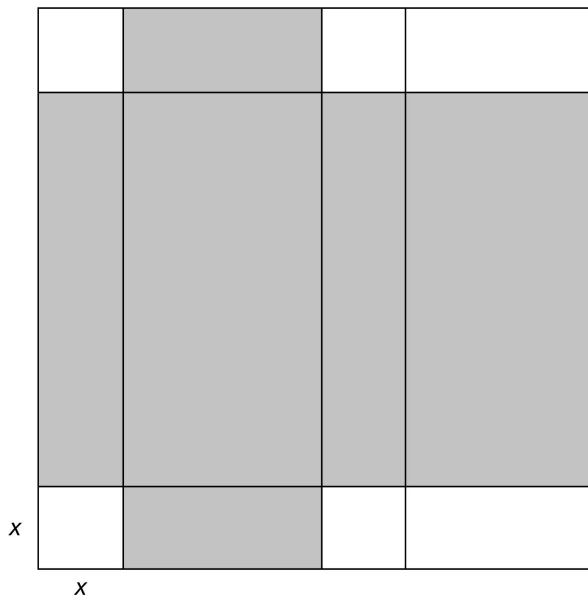
Evaluating m at the critical point and at each endpoint gives the following.

x	m
1	16
4	7
8	9

The minimum slope is $m = 7$, which occurs at $x = 4$.

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60.



The figure above represents a square sheet of cardboard with side length 20 inches. The sheet is cut and pieces are discarded. When the cardboard is folded, it becomes a rectangular box with a lid. The pattern for the rectangular box with a lid is shaded in the figure. Four squares with side length x and two rectangular regions are discarded from the cardboard. Which of the following statements is true? (The volume V of a rectangular box is given by $V = lwh$.)

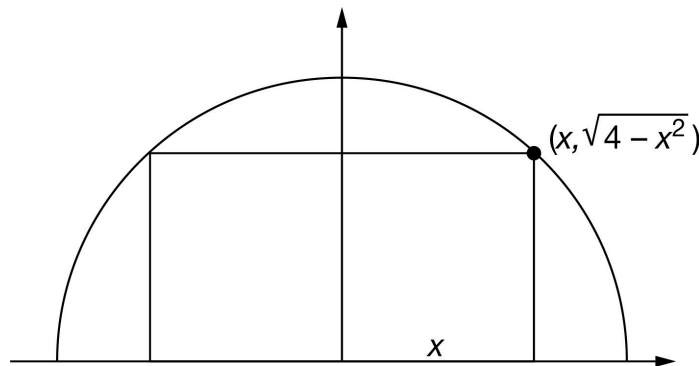
- (A) When $x = 10$ inches, the box has a minimum possible volume.
- (B) When $x = 10$ inches, the box has a maximum possible volume.
- (C) When $x = \frac{10}{3}$ inches, the box has a minimum possible volume.
- (D) When $x = \frac{10}{3}$ inches, the box has a maximum possible volume. ✓

Answer D

Correct. To find the volume of the rectangular box, the length, width, and height of the box must be identified. When the shaded region is folded to form the box, the height of the box is x inches. Since the side length of the cardboard sheet is 20 inches, the length of the box is $(20 - 2x)$ inches because each of the smaller squares has side length x . Label the width of the rectangular box as y inches. The side length of the cardboard sheet, in inches, can also be defined as $x + y + x + y = 2x + 2y = 20$, so $x + y = 10 \Rightarrow y = 10 - x$ is an expression for the width of the box in inches. Therefore, the volume of the box, in cubic inches, is $x(20 - 2x)(10 - x)$. To find the value of x that maximizes the volume, take the derivative of the expression. Since this is a polynomial, it might be easier to expand the polynomial before taking the derivative. This results in $2x^3 - 40x^2 + 200x$. Using the power rule, the derivative is $6x^2 - 80x + 200$. Factoring this quadratic polynomial as $2(3x^2 - 40x + 100) = 2(3x - 10)(x - 10) = 0 \Rightarrow x = \frac{10}{3}$ produces the maximum volume of the box.

5-9-11

61.



The figure above shows a rectangle inscribed in a semicircle with a radius of 2. The area of such a rectangle is given by $A(x) = 2x\sqrt{4 - x^2}$, where the width of the rectangle is $2x$. It can be shown that $A'(x) = \frac{-2x^2}{\sqrt{4 - x^2}} + 2\sqrt{4 - x^2}$ and A has critical values of -2 , $-\sqrt{2}$, $\sqrt{2}$, and 2 . It can also be shown that $A'(x)$ changes from positive to negative at $x = \sqrt{2}$. Which of the following statements is true?

- (A) The inscribed rectangle with maximum area has dimensions $\sqrt{2}$ by $\sqrt{2}$.
- (B) The inscribed rectangle with minimum area has dimensions $\sqrt{2}$ by $\sqrt{2}$.
- (C) The inscribed rectangle with maximum area has dimensions $2\sqrt{2}$ by $\sqrt{2}$. ✓
- (D) The inscribed rectangle with minimum area has dimensions $2\sqrt{2}$ by $\sqrt{2}$.

Answer C

Correct. The feasible domain of A is the open interval $0 < x < 2$. Because $x = \sqrt{2}$ is the only critical value in this interval and $A'(x)$ changes from positive to negative at $x = \sqrt{2}$, A attains an absolute maximum value at this x -value. The corresponding y -value is $y = \sqrt{2}$. Since the width of the rectangle is $2x$, the inscribed rectangle with maximum area has dimensions $2\sqrt{2}$ by $\sqrt{2}$.

5-9-11

62. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of mosquitoes in a field after a major rainfall is modeled by the function M defined by $M(t) = -t^3 + 12t^2 + 144t$, where t is the number of days after the rainfall ended and $0 \leq t \leq 18$.

- Using correct units, interpret the meaning of $M'(15) = -171$ in the context of the problem.
- Based on the model, what is the absolute maximum number of mosquitoes in the field over the time interval $0 \leq t \leq 18$? Justify your answer.
- For what values of t is the rate of change of the number of mosquitoes in the field increasing?
- For $0 \leq t \leq 18$, the number of bats in the field is modeled by the differentiable function B , where B is a function of the number of mosquitoes in the field. Based on the models, write an expression for the rate of change of the number of bats in the field at time $t = a$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct interpretation of $M'(15) = -171$

Solution:

At time $t = 15$ days, the number of mosquitoes in the field is decreasing at a rate of 171 mosquitoes per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $M'(t)$ with a maximum of one computational error.

5-9-11

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $M'(t)$
- ☐ sets $M'(t) = 0$
- ☐ identifies $t = 12$ as a candidate
- ☐ answer with justification

Solution:

$$M'(t) = -3t^2 + 24t + 144 = -3(t^2 - 8t - 48) = -3(t - 12)(t + 4)$$

$$M'(t) = 0 \Rightarrow t = 12$$

Because $M'(t)$ changes from positive to negative at $t = 12$, M has a relative maximum at $t = 12$.

Because $t = 12$ is the only critical point of M on the interval $0 \leq t \leq 18$, that is the location of a relative maximum. It is also the location of the absolute maximum of M on the interval $0 \leq t \leq 18$.

$$M(12) = -12^3 + 12(12^2) + 144(12) = 144(12) = 1728$$

The absolute maximum number of mosquitoes in the field over the time interval $0 \leq t \leq 18$ is $M(12) = 1728$.

Part C

The third point may be earned based on a consistent answer using an incorrect $M''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ considers $M''(t)$
- ☐ expression for $M''(t)$

5-9-11

☐ answer**Solution:**

$$M''(t) = -6t + 24 = -6(t - 4)$$

$$M''(t) = -6(t - 4) > 0 \Rightarrow t < 4$$

The rate of change of the number of mosquitoes in the field is increasing for $0 \leq t \leq 4$.

Part D

The point is earned with $B'(M(a)) \cdot M'(a)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct expression.

Solution:

The number of bats in the field at time $t = a$ is given by $B(M(a))$.

Therefore, the rate of change of the number of bats in the field at time $t = a$ is given by

$$B'(M(a)) \cdot M'(a) = B'(-a^3 + 12a^2 + 144a) \cdot (-3a^2 + 24a + 144).$$

5-9-11

63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of fish in a small bay is modeled by the function F defined by $F(t) = 10(t^3 - 12t^2 + 45t + 100)$, where t is measured in days and $0 \leq t \leq 8$.

- Using correct units, interpret the meaning of $F'(4) = -30$ in the context of the problem.
- Based on the model, what is the absolute minimum number of fish in the bay over the time interval $0 \leq t \leq 8$? Justify your answer.
- For what values of t , $0 \leq t \leq 8$, is the rate of change of the number of fish in the bay decreasing?
- For $0 \leq t \leq 8$, the number of pelicans flying near the bay is modeled by the differentiable function P , where P is a function of the number of fish in the bay. Based on the models, write an expression for the rate of change of the number of pelicans flying near the bay at time $t = c$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1
---	---

The student response accurately includes a correct interpretation of $F'(4) = -30$.

Solution:

At time $t = 4$ days, the number of fish in the bay is decreasing at a rate of 30 fish per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $F'(t)$ with a maximum of one computational error.

5-9-11

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $F'(t)$
- ☐ sets $F'(t) = 0$
- ☐ identifies $t = 5$ as a candidate
- ☐ answer with justification

Solution:

$$F'(t) = 10(3t^2 - 24t + 45) = 10 \cdot 3(t^2 - 8t + 15) = 30(t - 3)(t - 5)$$

$$F'(t) = 0 \Rightarrow t = 3 \text{ and } t = 5$$

Because $F'(t)$ changes from positive to negative at $t = 3$ and from negative to positive at $t = 5$, consider $t = 0$ and $t = 5$ for the location of the absolute minimum. F has a relative minimum at $t = 5$.

$$F(0) = 1000$$

$$F(5) = 10(5^3 - 12(5)^2 + 45(5) + 100) = 1500$$

The absolute minimum number of fish in the bay over the time interval

$$0 \leq t \leq 8 \text{ is } F(0) = 1000.$$

Part C

The third point may be earned based on a consistent answer using an incorrect $F''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

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- ☐ considers $F''(t)$
- ☐ expression for $F''(t)$
- ☐ answer

Solution:

$$F''(t) = 10(6t - 24) = 60(t - 4)$$

$$F''(t) = 60(t - 4) < 0 \Rightarrow t < 4$$

The rate of change of the number of fish in the bay is decreasing for $0 \leq t \leq 4$.

Part D

The point is earned with $P'(F(c)) \cdot F'(c)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct expression.

Solution:

The number of pelicans flying near the bay at time $t = c$ is given by $P(F(c))$.

Therefore, the rate of change of the number of pelicans flying near the bay at time $t = c$ is given by

$$P'(F(c)) \cdot F'(c) = P'(10(c^3 - 12c^2 + 45c + 100)) \cdot (10(3c^2 - 24c + 45)).$$

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64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	5	8	10
$G(t)$ (thousands of gallons)	22	18.5	19.5	22	24

A gasoline storage tank is filled by a pipeline from a refinery. At the same time, gasoline flows from the tank into trucks that will make deliveries to gasoline stations. The amount of gasoline in the storage tank at time t is given by the twice-differentiable function G , where t is measured in hours and $0 \leq t \leq 10$. Values of $G(t)$, in thousands of gallons, at selected times t are given in the table above. It is known that $G''(t) > 0$ for $5 \leq t < 10$.

- Use the data in the table to estimate the rate of change of the amount of gasoline in the storage tank at time $t = 3$ hours. Show the computations that lead to your answer. Indicate units of measure.
- For $0 < t < 10$, is there a time t at which $G(t)$ is increasing at a rate of 0.2 thousand gallons per hour? Justify your answer.
- It is known that $G'(5) = 0.5$. Use the locally linear approximation for G at time $t = 5$ to approximate the amount of gasoline in the storage tank at time $t = 7$. Is this approximation an overestimate or an underestimate for the actual amount of gasoline in the storage tank at time $t = 7$? Give a reason for your answer.
- The rate at which gasoline flows out of the storage tank into trucks at time t can be modeled by the function R defined by $R(t) = \frac{100t}{t^2+4}$, where t is measured in hours, and $R(t)$ is measured in thousands of gallons. Based on the model, at what time t , for $0 \leq t \leq 10$, is the rate at which gasoline flows out of the storage tank an absolute maximum? Justify your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct approximation with units.

Solution:

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$G'(3) \approx \frac{G(5)-G(1)}{5-1} = \frac{19.5-18.5}{5-1} = \frac{1}{4} = 0.25$ thousand gallons per hour (which is equivalent to 250 gallons per hour)

Part B

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{G(10)-G(0)}{10-0}$
- ☐ justification using Mean Value Theorem

Solution:

G is twice differentiable. $\Rightarrow G$ is differentiable. $\Rightarrow G$ is continuous.

$$\frac{G(10)-G(0)}{10-0} = \frac{24-22}{10-0} = \frac{2}{10} = 0.2 \text{ thousand gallons per hour (which is equivalent to 200 gallons per hour)}$$

Therefore, by the Mean Value Theorem, there is a time t , for $0 < t < 10$, at which $G'(t) = 0.2$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation of $G(7)$
- ☐ underestimate with reason

Solution:

$$G(7) \approx G(5) + G'(5)(7-5) = 19.5 + (0.5)(2) = 20.5 \text{ thousand gallons}$$

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$G''(t) > 0$ for $5 \leq t \leq 7$, and so the graph of G is concave up for $5 \leq t \leq 7$. Therefore, the approximation for $G(7)$ using the locally linear approximation for G at time $t = 5$ is an underestimate for the actual amount.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $R'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $R'(t)$
- ☐ sets $R'(t) = 0$
- ☐ identifies $t = 2$ as a candidate
- ☐ answer with justification

Solution:

$$R'(t) = \frac{100(t^2+4) - 100t(2t)}{(t^2+4)^2} = \frac{400 - 100t^2}{(t^2+4)^2} = \frac{100(4-t^2)}{(t^2+4)^2} = 0$$

$$\Rightarrow 4 - t^2 = 0 \Rightarrow t = 2$$

Because $R'(t)$ changes from positive to negative at $t = 2$, R has a relative maximum at $t = 2$. R is increasing on $0 \leq t \leq 2$ and R is decreasing on $2 \leq t \leq 10$. Since $t = 2$ is the only critical point of R on the interval $0 \leq t \leq 10$ that is the location of a relative maximum, it is also the location of the absolute maximum of R on the interval $0 \leq t \leq 10$.

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65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	5	15	30	35
$A(t)$ (gallons)	10	18	25	16	8

The number of gallons of olive oil in a tank at time t is given by the twice-differentiable function A , where t is measured in hours and $0 \leq t \leq 35$. Values of $A(t)$ at selected times t are given in the table above.

- Use the data in the table to estimate the rate at which the number of gallons of olive oil in the tank is changing at time $t = 10$ hours. Show the computations that lead to your answer. Indicate units of measure.
- For $0 \leq t \leq 30$, is there a time t at which $A'(t) = \frac{1}{5}$? Justify your answer.
- The number of gallons of olive oil in the tank at time t is also modeled by the function G defined by $G(t) = 5t - \frac{2}{3}(t+9)^{\frac{3}{2}} + 28$, where t is measured in hours and $0 \leq t \leq 35$. Based on the model, at what time t , for $0 \leq t \leq 35$, is the number of gallons of olive oil in the tank an absolute maximum? Justify your answer.
- For the function G defined in part (c), the locally linear approximation near $t = 7$ is used to approximate $G(8)$. Is this approximation an overestimate or an underestimate for the value of $G(8)$? Give a reason for your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ approximation with units

Solution:

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$$A'(10) \approx \frac{A(15) - A(5)}{15 - 5} = \frac{25 - 18}{15 - 5} = \frac{7}{5} \text{ gallons per hour}$$

Part B

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{A(30) - A(0)}{30 - 0}$
- ☐ justification using Mean Value Theorem

Solution:

A is twice differentiable. $\Rightarrow A$ is differentiable. $\Rightarrow A$ is continuous.

$$\frac{A(30) - A(0)}{30 - 0} = \frac{16 - 10}{30 - 0} = \frac{6}{30} = \frac{1}{5}$$

Therefore, by the Mean Value Theorem, there is a time t , for $0 \leq t \leq 30$, at which $A'(t) = \frac{1}{5}$.

Part C

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $G'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $G'(t)$

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- ☐ sets $G'(t) = 0$
- ☐ identifies $t = 16$ as a candidate
- ☐ answer with justification

Solution:

$$G'(t) = 5 - (t + 9)^{\frac{1}{2}} = 0 \Rightarrow (t + 9)^{\frac{1}{2}} = 5$$

$$\Rightarrow t + 9 = 25 \Rightarrow t = 16$$

Because $G'(t)$ changes from positive to negative at $t = 16$, G has a relative maximum at $t = 16$.

Since $t = 16$ is the only critical point of G on the interval $0 \leq t \leq 35$ that is the location of a relative maximum, it is also the location of the absolute maximum of G on the interval $0 \leq t \leq 35$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $G''(t)$
- ☐ overestimate with reason

Solution:

$$G''(t) = -\frac{1}{2}(t + 9)^{-\frac{1}{2}}$$

$G''(t) < 0$ for $7 \leq t \leq 8$, and so the graph of G is concave down for $7 \leq t \leq 8$. Therefore, the approximation of $G(8)$ using the locally linear approximation of G near $t = 7$ is an overestimate.

5-9-11

66. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Fish enter a lake at a rate modeled by the function E given by $E(t) = 20 + 15 \sin\left(\frac{\pi t}{6}\right)$. Fish leave the lake at a rate modeled by the function L given by $L(t) = 4 + 2^{0.1t^2}$. Both $E(t)$ and $L(t)$ are measured in fish per hour, and t is measured in hours since midnight ($t = 0$).

- How many fish enter the lake over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$)? Give your answer to the nearest whole number.
- What is the average number of fish that leave the lake per hour over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$)?
- At what time t , for $0 \leq t \leq 8$, is the greatest number of fish in the lake? Justify your answer.
- Is the rate of change in the number of fish in the lake increasing or decreasing at 5 A.M. ($t = 5$)? Explain your reasoning.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point requires a correct integrand and limits.

The second point requires the first point to be earned. The second point is earned with an answer of 153, 154, or any noninteger answer that is accurate to the number of places presented after the decimal point (up to the first 3). An answer between 101 and 102, inclusive, might indicate that the response uses degree mode rather than radian mode. In this case, the second point is not earned, but the remaining points for the question may be earned for consistent answers using degree mode.



0	1	2
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5-9-11

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\int_0^5 E(t) dt = 153.457690$$

To the nearest whole number, 153 fish enter the lake from midnight to 5 A.M.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a correct integrand and limits. The $\frac{1}{5}$ is considered for the second point. If this integral is computed in Part A of the question, that number can be used in place of the integral to earn the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\frac{1}{5-0} \int_0^5 L(t) dt = 6.059038$$

The average number of fish that leave the lake per hour from midnight to 5 A.M. is 6.059 fish per hour.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned through analytical, graphical, or verbal expressions to show where $E(t) - L(t) = 0$, where $E(t) - L(t)$ changes sign, or where $E(t)$ and $L(t)$ intersect.

The second point is earned with the correct value of t .

The third point requires a global argument for the sign of $E(t) - L(t)$ rather than a local argument.

5-9-11

Another approach uses the Candidates Test. Let $A(t)$ be the change in the number of fish in the lake from midnight to t hours after midnight.

$$A(t) = \int_0^t (E(s) - L(s)) \, ds$$

$$A'(t) = E(t) - L(t) = 0 \Rightarrow t = C = 6.20356$$

t	$A(t)$
0	0
C	135.01492
8	80.91908

Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ sets $E(t) - L(t) = 0$
- ☐ answer
- ☐ justification

Solution:

The rate of change in the number of fish in the lake at time t is given by $E(t) - L(t)$.

$$E(t) - L(t) = 0 \Rightarrow t = 6.20356$$

$E(t) - L(t) > 0$ for $0 \leq t < 6.20356$, and $E(t) - L(t) < 0$ for $6.20356 < t \leq 8$. Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point requires an attempt to use $E'(5)$ and $L'(5)$. Incorrect symbolic differentiation for both evaluated at 5 earns the point. If values are reported without symbolic work, at least one must be correct to earn the point. These values need not be reported to three places after the decimal point.

The second point requires a direct connection between $E'(5)$ and $L'(5)$. The point is not earned if reported values are incorrect. The numerical value of $E'(5) - L'(5)$ is not required, but the response must indicate that this value is negative.

5-9-11



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ considers $E'(5)$ and $L'(5)$
- ☐ answer with explanation

Solution:

$$E'(5) - L'(5) = -10.7228 < 0$$

Because $E'(5) - L'(5) < 0$, the rate of change in the number of fish is decreasing at time $t = 5$.

5-9-11

67.  Oil is leaking from a pipeline on the surface of a lake and forms an oil slick whose volume increases at a constant rate of 2000 cubic centimeters per minute. The oil slick takes the form of a right circular cylinder with both its radius and height changing with time. (Note: The volume V of a right circular cylinder with radius r and height h is given by $V = \pi r^2 h$.)
- (a) At the instant when the radius of the oil slick is 100 centimeters and the height is 0.5 centimeter, the radius is increasing at the rate of 2.5 centimeters per minute. At this instant, what is the rate of change of the height of the oil slick with respect to time, in centimeters per minute?
- (b) A recovery device arrives on the scene and begins removing oil. The rate at which oil is removed is $R(t) = 400\sqrt{t}$ cubic centimeters per minute, where t is the time in minutes since the device began working. Oil continues to leak at the rate of 2000 cubic centimeters per minute. Find the time t when the oil slick reaches its maximum volume. Justify your answer.
- (c) By the time the recovery device began removing oil, 60,000 cubic centimeters of oil had already leaked. Write, but do not evaluate, an expression involving an integral that gives the volume of oil at the time found in part (b).

Part A

1 point earned for $\frac{dV}{dt} = 2000$ and $\frac{dr}{dt} = 2.5$

When $r = 100$ cm and $h = 0.5$ cm, $\frac{dV}{dt} = 2000 \text{ cm}^3/\text{min}$ and $\frac{dr}{dt} = 2.5 \text{ cm}/\text{min}$.

2 point earned for the correct expression of $\frac{dV}{dt}$

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

1 point earned for the correct answer

$$2000 = 2\pi(100)(2.5)(0.5) + \pi(100)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.038 \text{ or } 0.039 \text{ cm}/\text{min}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point earned for $\frac{dV}{dt} = 2000$ and $\frac{dr}{dt} = 2.5$

5-9-11

When $r = 100$ cm and $h = 0.5$ cm, $\frac{dV}{dt} = 2000$ cm³/min and $\frac{dr}{dt} = 2.5$ cm/min.

2 point earned for the correct expression of $\frac{dV}{dt}$

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

1 point earned for the correct answer

$$2000 = 2\pi(100)(2.5)(0.5) + \pi(100)^2 \frac{dh}{dt}$$

$$\frac{dh}{dt} = 0.038 \text{ or } 0.039 \text{ cm/min}$$

Part B

1 point earned for $R(t) = 2000$

$$\frac{dV}{dt} = 2000 - R(t), \text{ so } \frac{dV}{dt} = 0 \text{ when } R(t) = 2000.$$

1 point earned for correct answer

This occurs when $t = 25$ minutes.

1 point earned for the justification

Since $\frac{dV}{dt} > 0$ for $0 < t < 25$ and $\frac{dV}{dt} < 0$ for $t > 25$, the oil slick reaches its maximum volume 25 minutes after the device begins working.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point earned for $R(t) = 2000$

$$\frac{dV}{dt} = 2000 - R(t), \text{ so } \frac{dV}{dt} = 0 \text{ when } R(t) = 2000.$$

1 point earned for correct answer

This occurs when $t = 25$ minutes.

1 point earned for the justification

5-9-11

Since $\frac{dV}{dt} > 0$ for $0 < t < 25$ and $\frac{dV}{dt} < 0$ for $t > 25$, the oil slick reaches its maximum volume 25 minutes after the device begins working.

Part C

1 point earned for the correct limits and initial condition

1 point earned for the integrand

The volume of oil, in cm^3 , in the slick at time $t = 25$ minutes is given by $60,000 + \int_0^{25} (2000 - R(t))dt$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point earned for the correct limits and initial condition

1 point earned for the integrand

The volume of oil, in cm^3 , in the slick at time $t = 25$ minutes is given by $60,000 + \int_0^{25} (2000 - R(t))dt$.

5-9-11

68. On what interval is the graph of $y = 6x^2 - x^3 - 5$ increasing and concave up?

- (A) $(0, 1)$ only
(B) $(0, 2)$ ✓
(C) $(1, 4)$
(D) $(2, 4)$ only

Answer B

Correct. Let $g(x) = 6x^2 - x^3 - 5$. The graph of $y = g(x)$ is increasing when $g'(x) > 0$ and concave up when $g''(x) > 0$.

$$g'(x) = 12x - 3x^2 = 3x(4 - x)$$

The graph of g' is a parabola opening down with zeros at $x = 0$ and $x = 4$, so g' is positive for $0 < x < 4$.

$$g''(x) = 12 - 6x = 6(2 - x)$$

This is a line with negative slope and a zero at $x = 2$. Therefore $g''(x) > 0$ for $x < 2$. The intersection of the intervals $(0, 4)$ and $(-\infty, 2)$ is $(0, 2)$.

69. A rectangle has its upper two vertices on the graph of $y = 12 - \frac{x^2}{3}$ and its lower two vertices on the x -axis. What is the maximum possible area of the rectangle?

- (A) $216 - 72\sqrt{5}$
(B) $32\sqrt{3}$ ✓
(C) 96
(D) 144

Answer B

Correct. The graph of $y = 12 - \frac{x^2}{3}$ is an inverted parabola symmetric to the y -axis. Let the lower two vertices of the rectangle be at x and $-x$ on the x -axis with $0 \leq x \leq 6$. The area of the rectangle is

$$A(x) = 2x \left(12 - \frac{x^2}{3} \right) = 24x - \frac{2}{3}x^3.$$

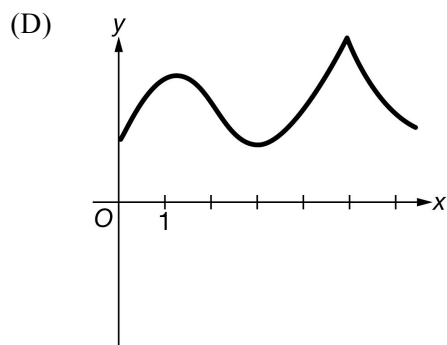
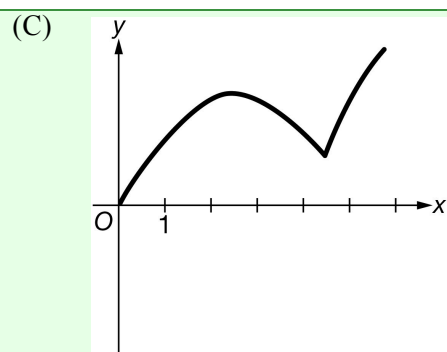
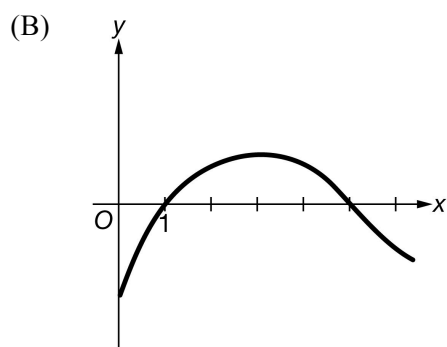
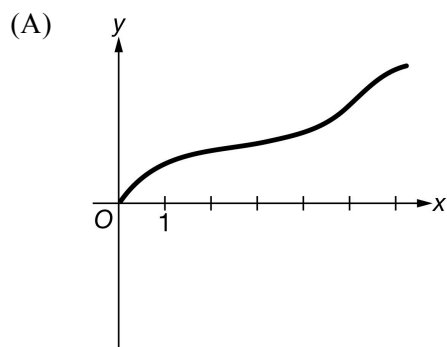
$$A'(x) = 24 - 2x^2 = 0 \Rightarrow x^2 = 12 \Rightarrow x = 2\sqrt{3}$$

$$A''(x) = -4x \Rightarrow A''(2\sqrt{3}) < 0$$

Therefore, $x = 2\sqrt{3}$ is a local maximum, and since it is the only critical point, it is a global maximum. The maximum possible area of the rectangle is therefore $A(2\sqrt{3}) = 48\sqrt{3} - 16\sqrt{3} = 32\sqrt{3}$.

5-9-11

70. The function f is differentiable and increasing on the interval $0 \leq x \leq 6$, and the graph of f has exactly two points of inflection on this interval. Which of the following could be the graph of f' , the derivative of f ?

**Answer C**

Correct. Because f is increasing on the interval $(0, 6)$, its derivative should satisfy $f'(x) \geq 0$ for $0 \leq x \leq 6$.

Because the graph of f has exactly two points of inflection, the concavity of f must change twice, and therefore the

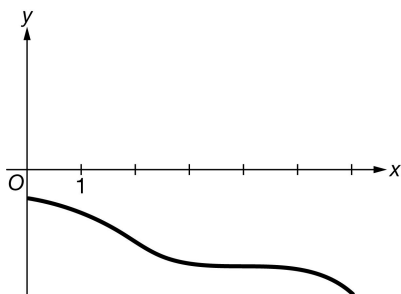
5-9-11

graph of f' must change from increasing to decreasing, or decreasing to increasing, twice on the interval $(0, 6)$. The relative maxima and relative minima of the graph of the derivative indicate points of inflection of the graph of f . Both of those conditions are satisfied by this graph, so this could be the graph of f' .

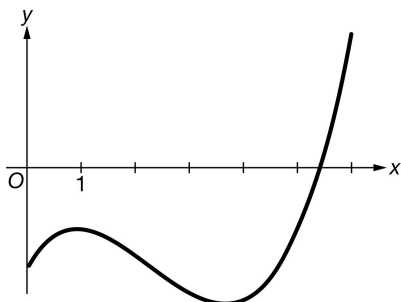
5-9-11

71. The function f is differentiable and decreasing on the interval $0 \leq x \leq 6$, and the graph of f has exactly two points of inflection on this interval. Which of the following could be the graph of f' , the derivative of f ?

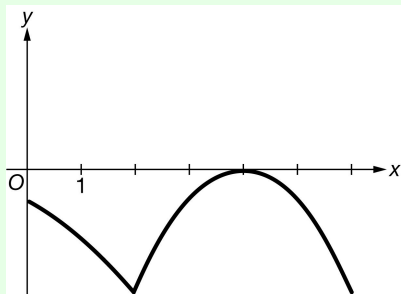
(A)



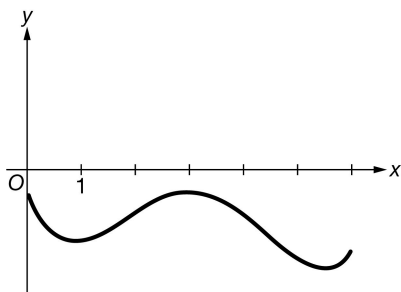
(B)



(C)



(D)



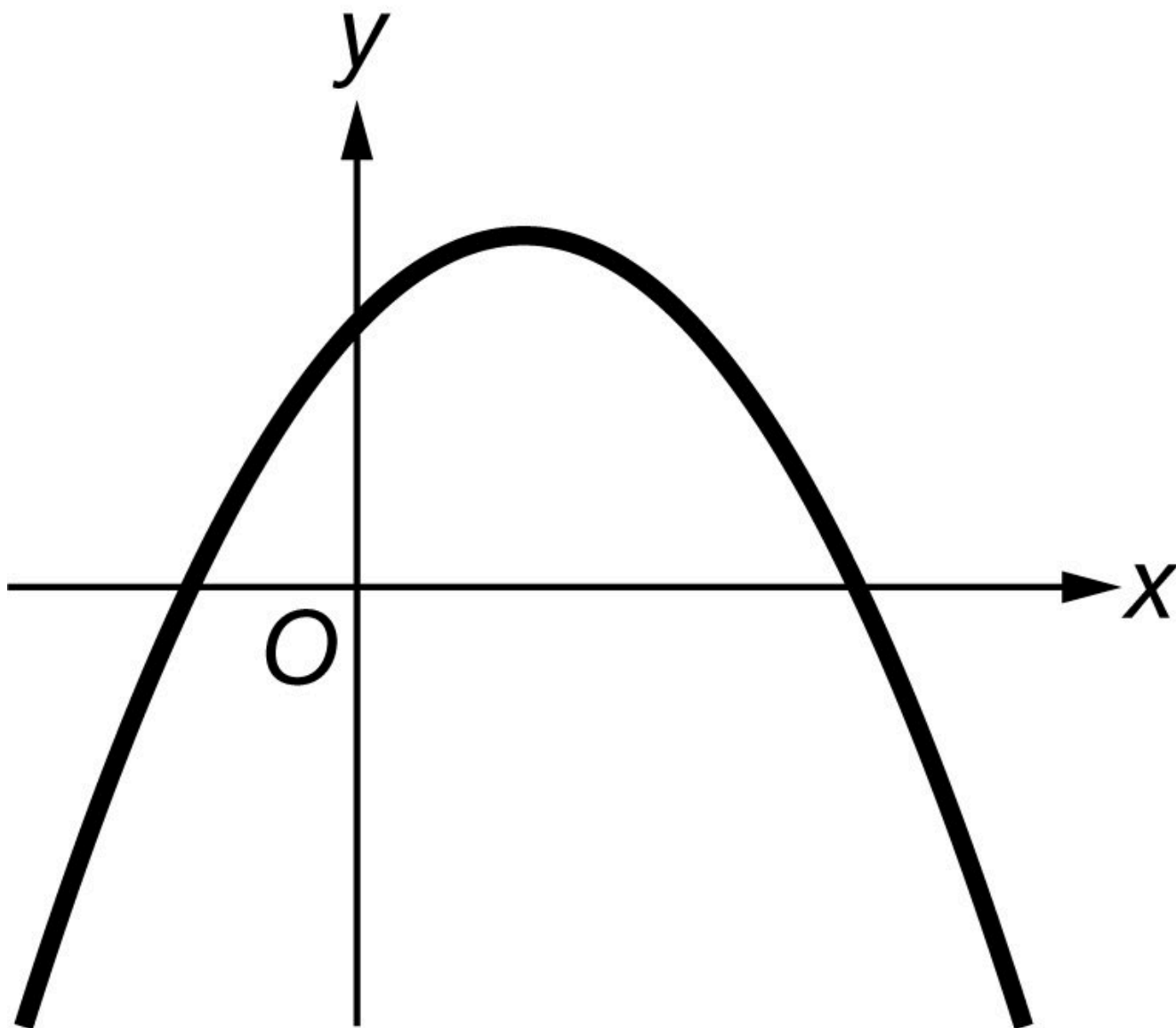
Answer C

Correct. Because f is decreasing on the interval $(0, 6)$, its derivative should satisfy $f'(x) \leq 0$ for $0 \leq x \leq 6$.

Because the graph of f has exactly two points of inflection, the concavity of f must change twice, and therefore the graph of f' must change directions twice on the interval $(0, 6)$. Both of those conditions are satisfied by this graph, so this could be the graph of f' .

5-9-11

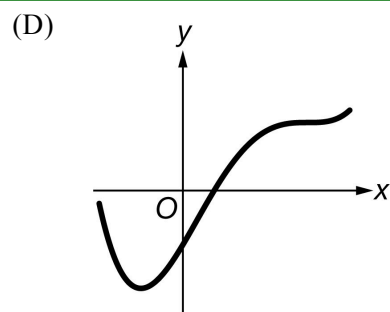
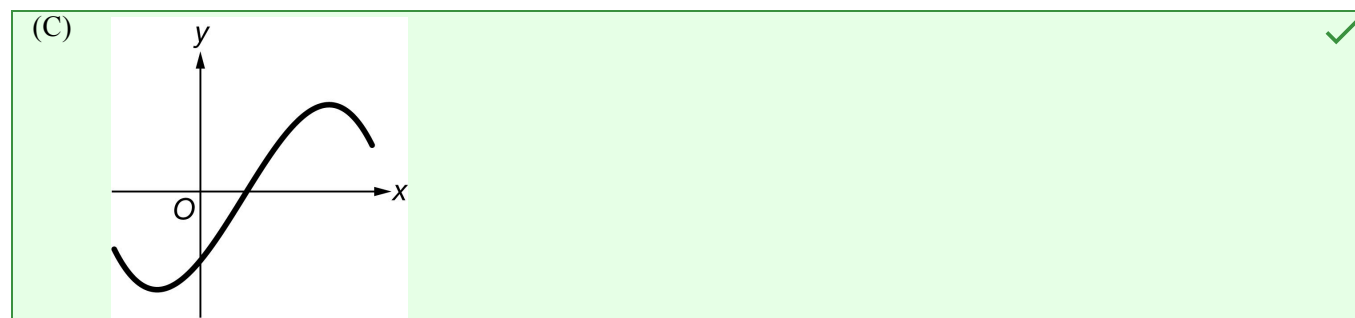
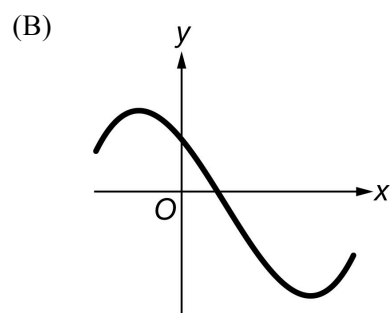
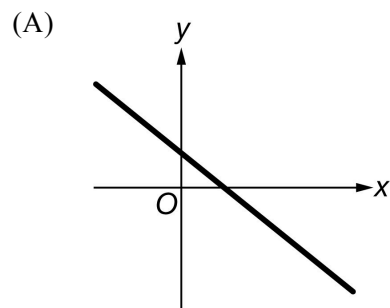
72.



Graph of f'

The graph of f' , the derivative of the function f , is shown above. Which of the following could be the graph of f ?

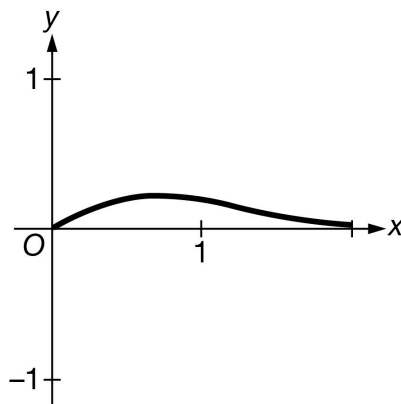
5-9-11

**Answer C**

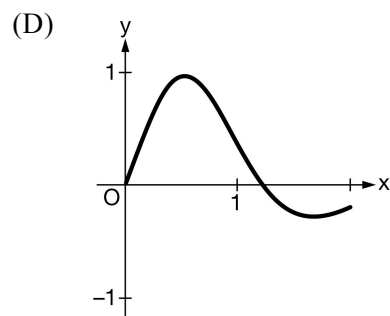
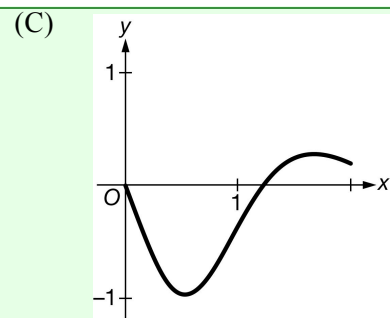
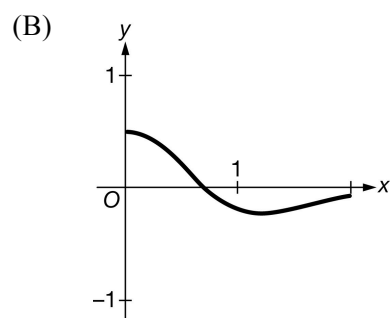
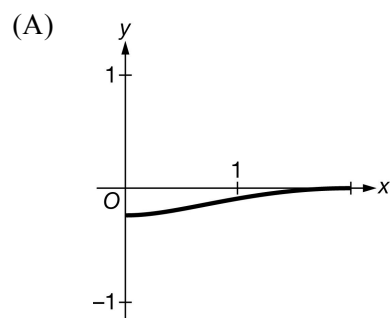
Correct. The graph of f' has two zeros indicating that f has two critical points. The first is a local minimum where f' changes from negative to positive. The second is a local maximum where f' changes from positive to negative. This graph matches that behavior and therefore could be the graph of f .

5-9-11

73.

Graph of f

The graph of $y = f(x)$ is shown above. Which of the following could be the graph of $y = f''(x)$?

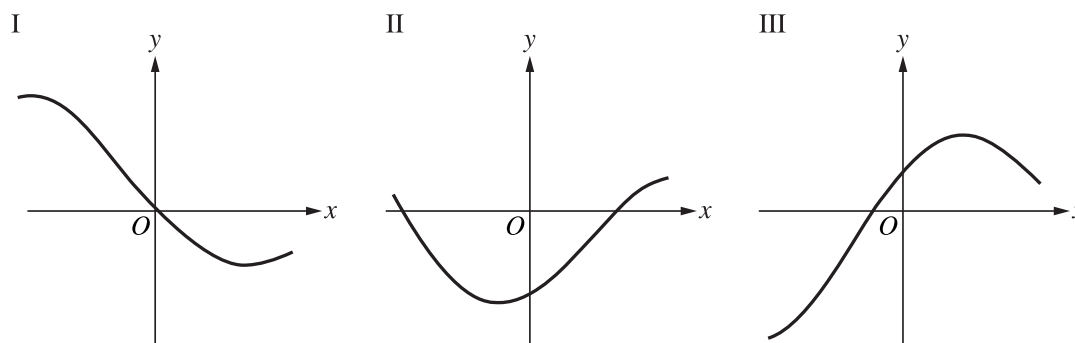


5-9-11

Answer C

Correct. The graph of f has a local maximum between 0 and 1, so it must be concave down around the local maximum. The graph of f appears to have a point of inflection slightly to the right of $x = 1$. Therefore the graph of f'' should be negative between 0 and 1 and have a zero slightly to the right of $x = 1$. This graph is the only one with these properties, so it could be the graph of $y = f''(x)$.

74.



Three graphs labeled I, II, and III are shown. One is the graph of f , one is the graph of f' , and one is the graph of f'' . Which of the following correctly identifies each of the three graphs?

- (A) The graph of f is I, the graph of f' is II, and the graph of f'' is III. ✓
- (B) The graph of f is I, the graph of f' is III, and the graph of f'' is II.
- (C) The graph of f is II, the graph of f' is III, and the graph of f'' is I.
- (D) The graph of f is III, the graph of f' is II, and the graph of f'' is I.

Answer A

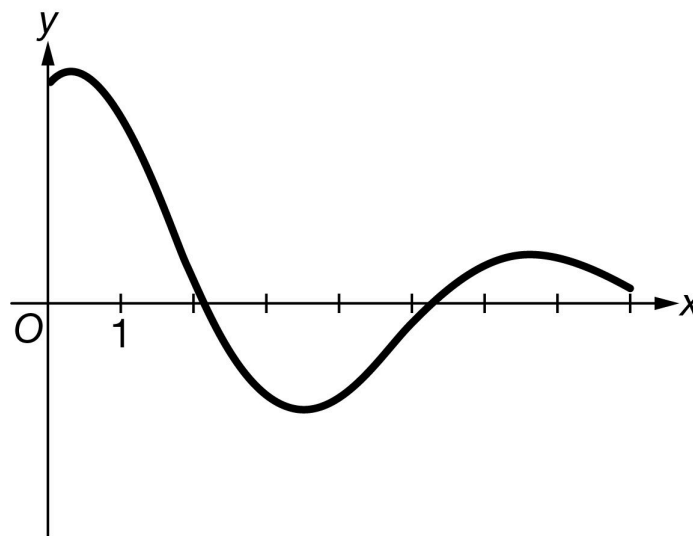
Correct. Graph I is decreasing over most of the interval shown, so its derivative must be negative over the corresponding part of the interval. Graph II is the only graph that is negative over the corresponding interval. In addition, graph I has a local maximum at an $x < 0$ and a local minimum at an $x > 0$. Graph II has zeros at the corresponding values of x where graph I changes from positive to negative and where graph I changes from negative to positive. So graph II could be the derivative of graph I.

Graph II has a local minimum just to the left of the origin, so its derivative must change from negative to positive at that location. Graph III is the only one with that property, so graph III could be the derivative of graph II. In addition, graph I has an apparent inflection point at the same location where graph III has a zero, so graph III could be the second derivative of graph I.

So I is the graph of f , II is the graph of f' , and III is the graph of f'' .

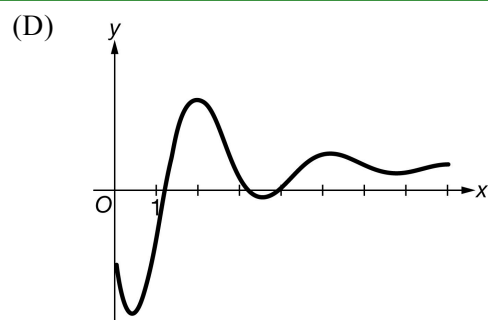
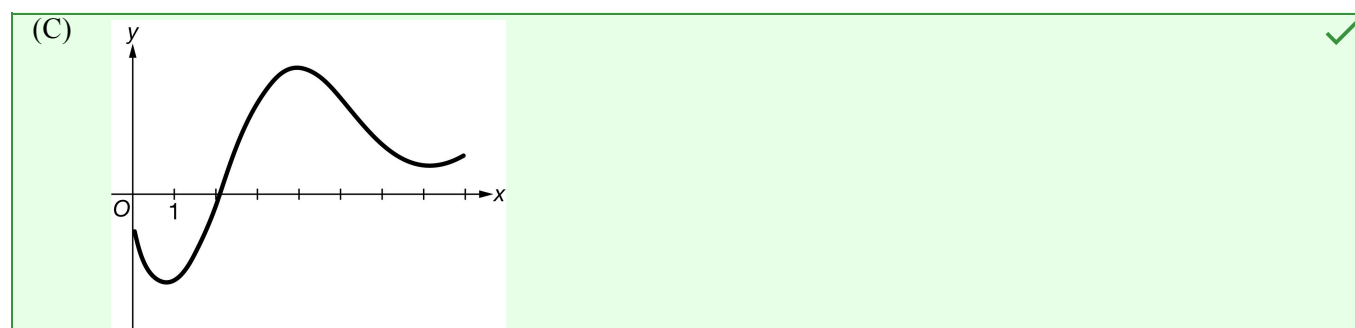
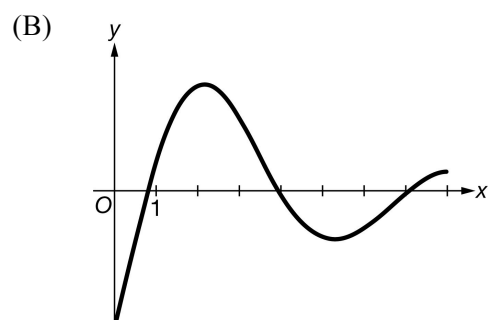
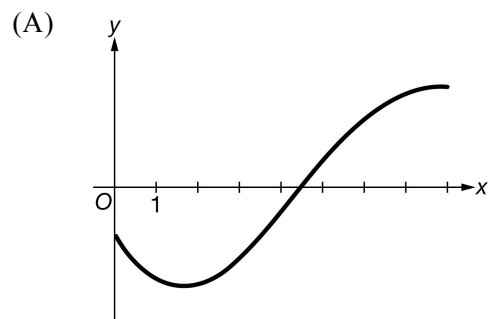
5-9-11

75.

Graph of f''

The graph of f'' , the second derivative of the function f , is shown above on the interval $0 \leq x \leq 8$. Which of the following could be the graph of f ?

5-9-11

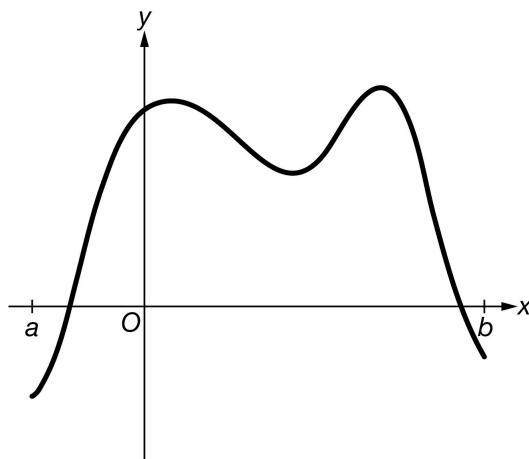


Answer C

Correct. The graph of f'' is positive until slightly past $x = 2$, then negative until slightly past $x = 5$, then positive again. This means that the graph of f should be concave up, then concave down, then concave up with two points of inflection at approximately $x = 2$ and $x = 5$. This graph demonstrates that behavior and could therefore be the graph of f .

5-9-11

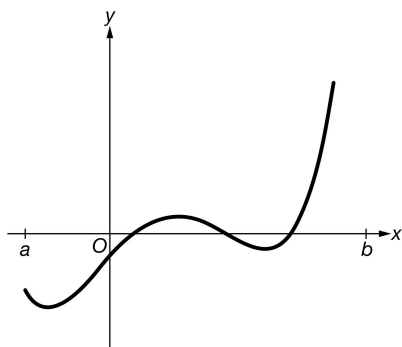
76.

Graph of f

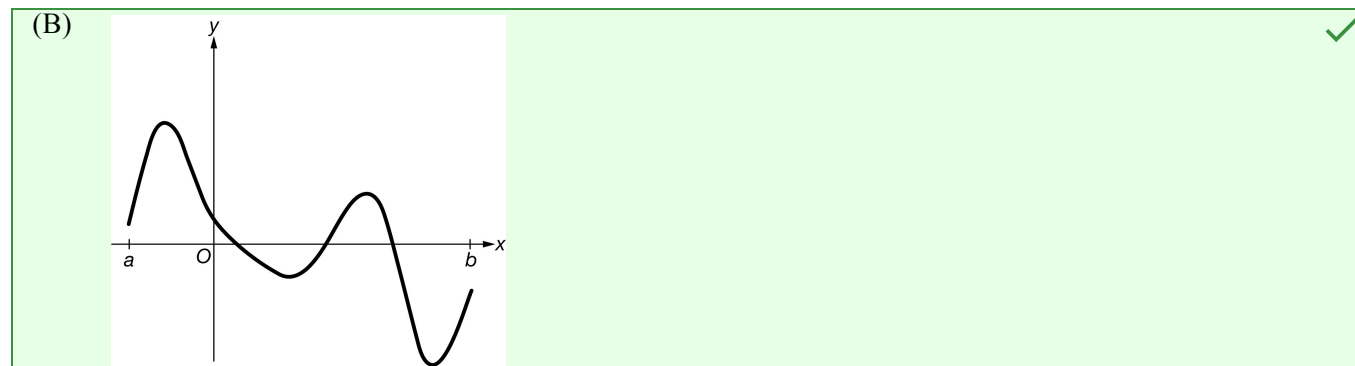
The figure above shows the graph of f on the interval $[a, b]$. Which of the following could be the graph of f' , the derivative of f , on the interval $[a, b]$?

5-9-11

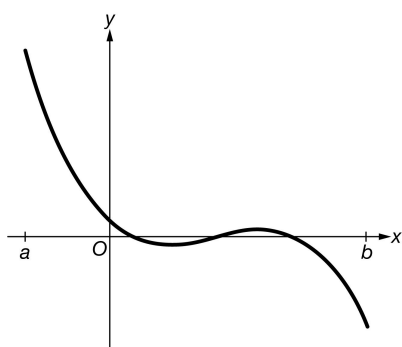
(A)



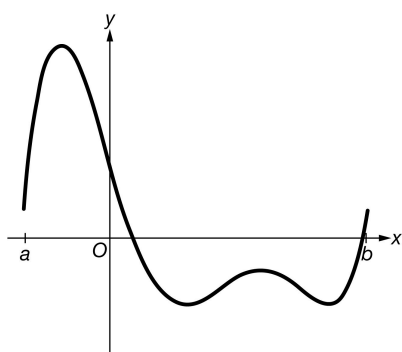
(B)



(C)



(D)

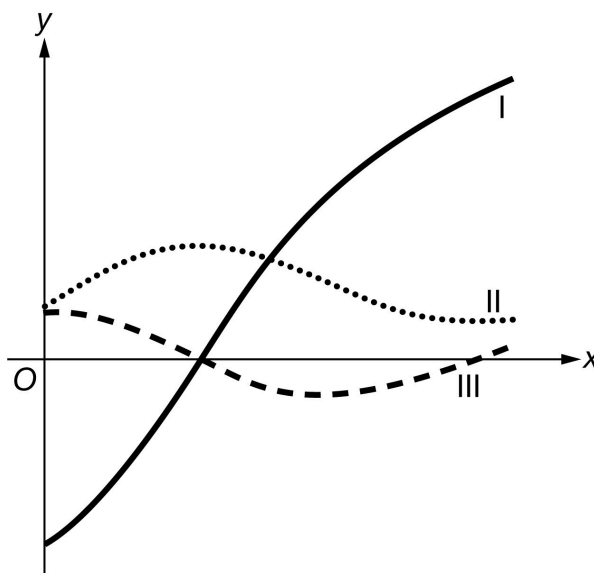
**Answer B**

Correct. The graph of f has a local maximum followed by a local minimum followed by a local maximum. Therefore the graph of f' should have three zeros where the graph goes from positive to negative, from negative to positive, and from positive to negative, respectively.

5-9-11

The graph of f also changes from concave up to concave down, then concave down to concave up, then concave up to concave down, then concave down to concave up. Therefore the graph of f' should have four relative extrema. This graph displays those properties on the interval $[a, b]$ and therefore could be the graph of f' .

77.



The figure above shows three graphs, labeled I, II, and III. One is the graph of f , one is the graph of f' , and one is the graph of f'' . Which of the following correctly identifies each of the three graphs?

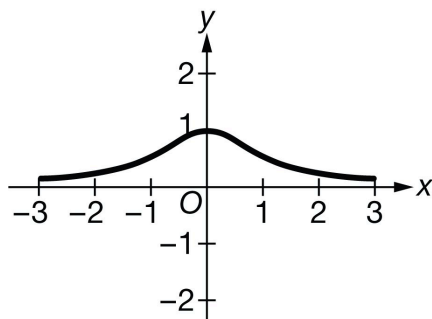
- (A) The graph of f is I, the graph of f' is II, and the graph of f'' is III. ✓
- (B) The graph of f is I, the graph of f' is III, and the graph of f'' is II.
- (C) The graph of f is II, the graph of f' is I, and the graph of f'' is III.
- (D) The graph of f is II, the graph of f' is III, and the graph of f'' is I.

Answer A

Correct. Graph I is increasing, so its derivative must be positive. Graph II is the only graph that is positive over the given interval, so graph II could be the derivative of graph I. Graph II has a local maximum, so its derivative must change from positive to negative at that location. Graph III has that property. In addition, graph II appears to have a local minimum where graph III changes from negative to positive. Therefore graph III is the derivative of graph II. So graph I is the graph of f , graph II is the graph of f' , and graph III is the graph of f'' .

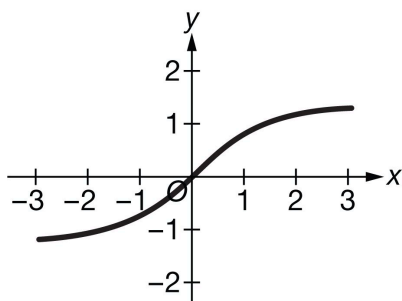
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78.

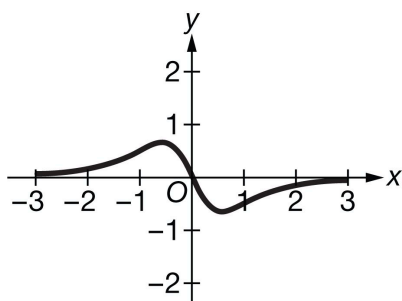
Graph of f

The graph of $y = f(x)$ is shown above. Which of the following could be the graph of $y = f''(x)$?

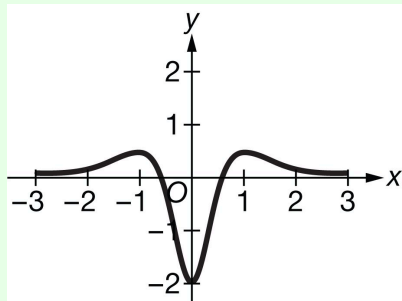
(A)



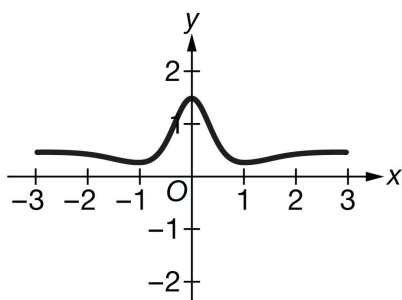
(B)



(C)



(D)

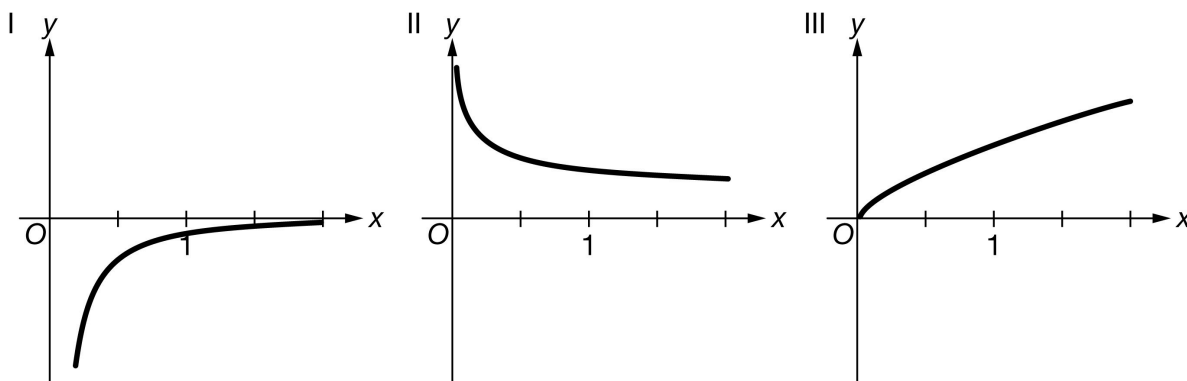


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Answer C

Correct. The graph of f has a local maximum at $x = 0$, so it must be concave down around the local maximum. The graph of f appears to have a point of inflection at a point in each of the intervals $(-1, 0)$ and $(0, 1)$; therefore, $f''(x)$ should have zeros in each of these intervals. The graph of f'' should change from positive to negative on $(-1, 0)$ and should change from negative to positive on $(0, 1)$. This graph is the only one with these properties, so it could be the graph of $f''(x)$.

79.



Three graphs labeled I, II, and III are shown above. One is the graph of f , one is the graph of f' , and one is the graph of f'' . Which of the following correctly identifies each of the three graphs?

(A)

f	f'	f''
I	II	III

(B)

f	f'	f''
II	III	II

(C)

f	f'	f''
II	II	II

(D)

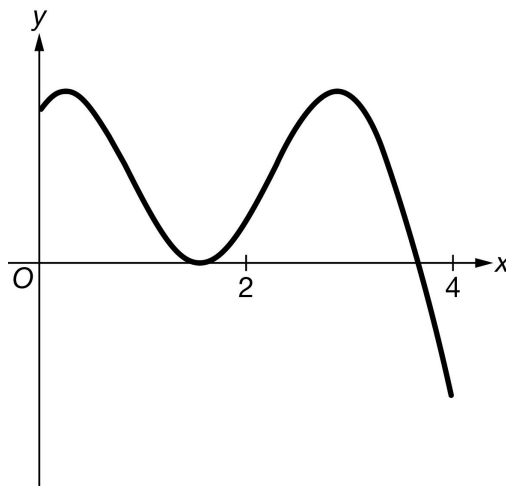
f	f'	f''
II	III	I

**Answer D**

Correct. Graph III is increasing and concave down for all $x > 0$. Therefore, the derivative of graph III should have positive values for all $x > 0$ and the second derivative should have negative values for all $x > 0$. Graph II is the derivative of graph III, and graph I is the second derivative of graph III.

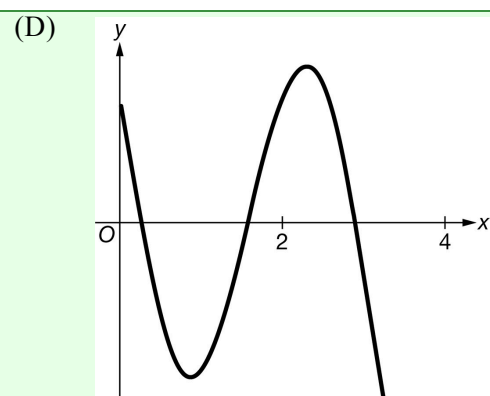
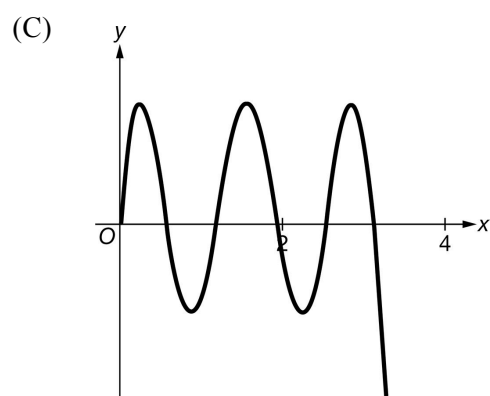
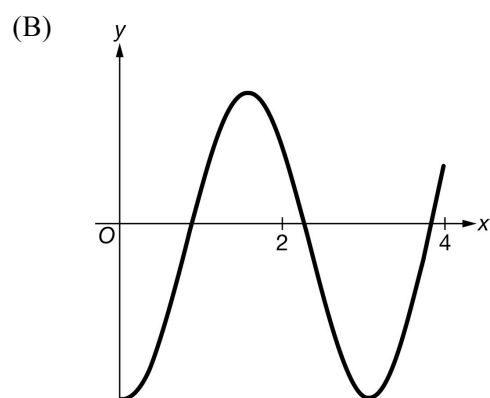
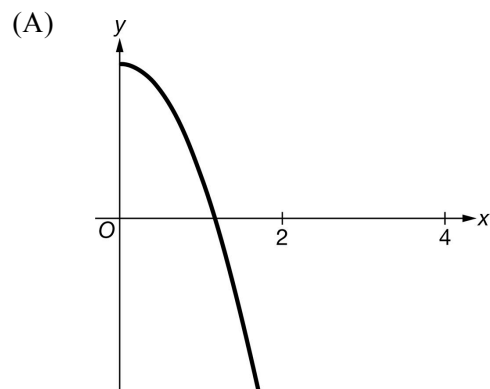
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80.

Graph of f

The figure above shows the graph of f on the interval $[0, 4]$. Which of the following could be the graph of f' , the derivative of f , on the interval $[0, 4]$?

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**Answer D**

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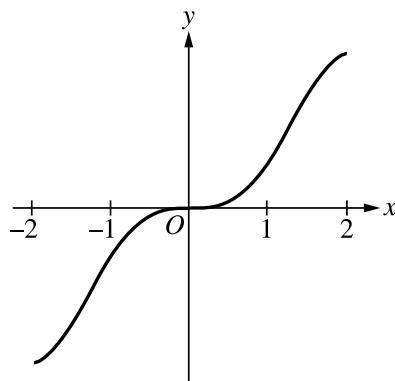
Correct. The graph of f has three extrema: a local maximum, followed by a local minimum and a local maximum. Therefore, the graph of f' should have three zeros where the graph goes from positive to negative, from negative to positive, and from positive to negative.

The graph of f changes from concave down to concave up to the left of 2 and changes from concave up to concave down to the right of 2. Therefore, the graph of f' should have two relative extrema at the points where the concavity of the graph of f changes.

This graph displays those properties on the interval $[0, 4]$ and therefore could be the graph of f' .

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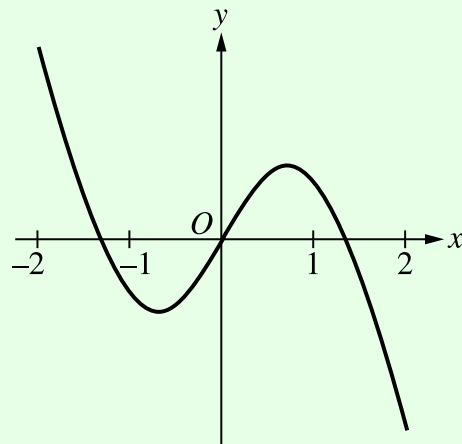
81.

Graph of f

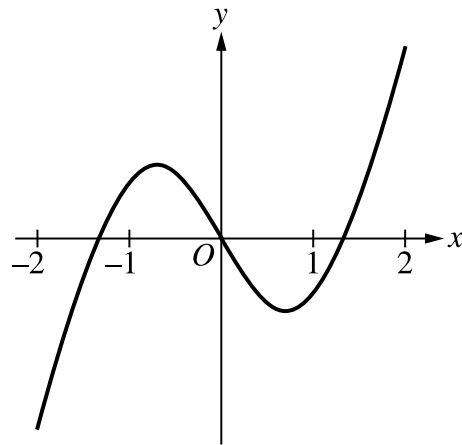
The preceding figure shows the graph of the twice-differentiable function f . Which of the following could be the graph of f'' , the second derivative of f ?

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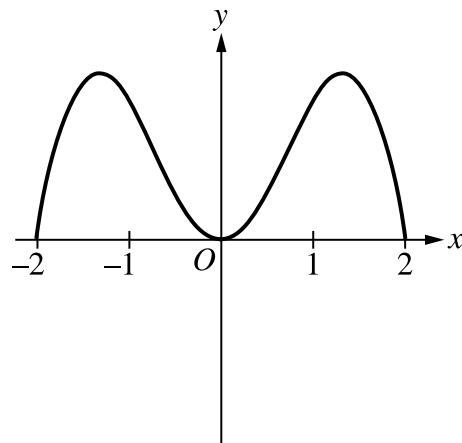
(A)



(B)

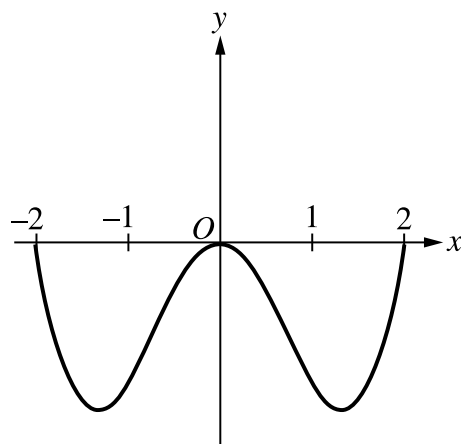


(C)



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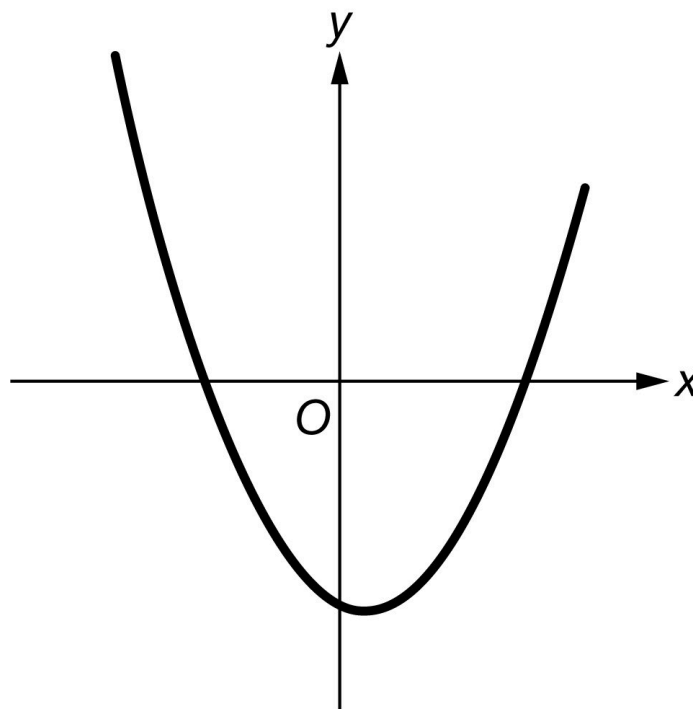
(D)

**Answer A**

Correct. The sign of the second derivative indicates the concavity of the graph of f . The graph of f starts out concave up, then changes to concave down just before $x = -1$. The graph changes back to concave up at the origin, then changes to concave down just after $x = 1$. Therefore the graph of f'' should start out positive, change to negative just before $x = -1$, change back to positive at the origin, and change to negative just after $x = 1$. This is the only graph for which those conditions hold.

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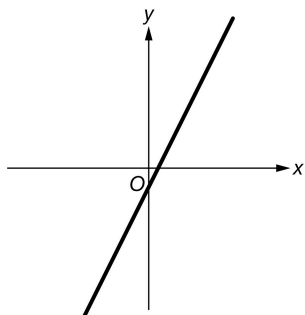
82.

Graph of f'

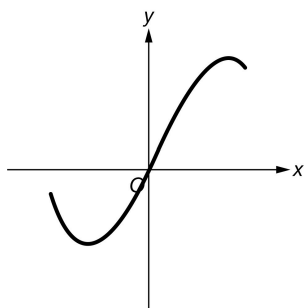
The graph of f' , the derivative of the function f , is shown above. Which of the following could be the graph of f ?

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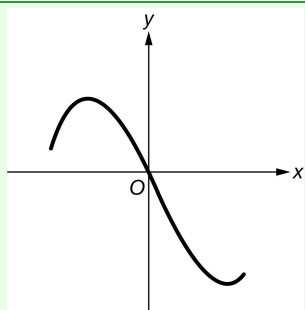
(A)



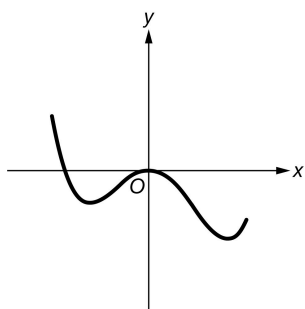
(B)



(C)



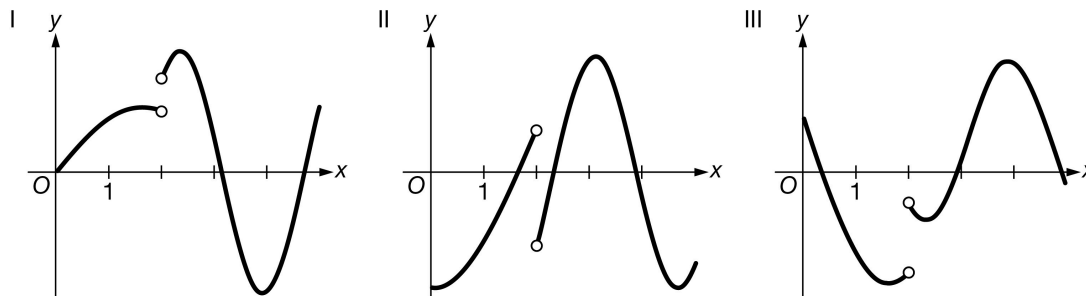
(D)

**Answer C**

Correct. The graph of f' has two zeros indicating that f has two critical points. The first is a local maximum where f' changes from positive to negative. The second is a local minimum where f' changes from negative to positive. This graph matches that behavior and therefore could be the graph of f .

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83.



Three graphs labeled I, II, and III are shown above. One is the graph of f , one is the graph of f' , and one is the graph of f'' . Which of the following correctly identifies each of the three graphs?

(A)

f	f'	f''
I	II	III

(B)

f	f'	f''
II	III	I

(C)

f	f'	f''
III	I	II

(D)

f	f'	f''
III	II	I

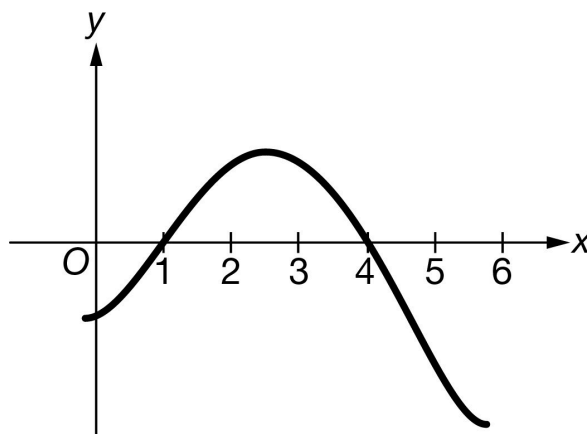


Answer D

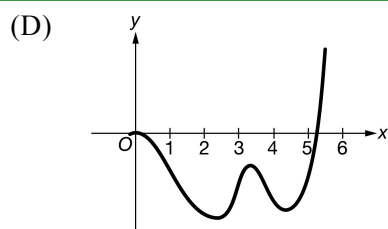
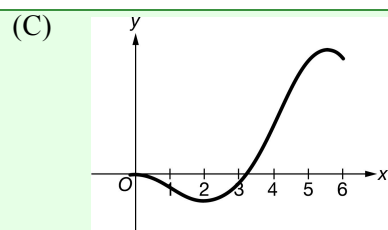
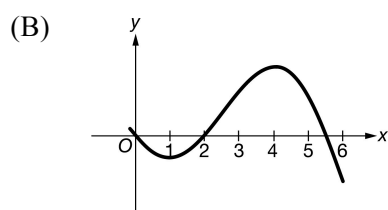
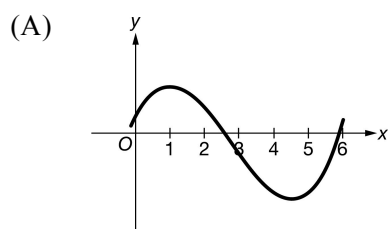
Correct. Graph II is the derivative of graph III. Graph III has two local minimums and one local maximum. Graph II has two zeros where the graph changes from negative to positive, and one zero where the graph changes from positive to negative. The location of the zeros in graph II matches the locations of the local extrema in graph III. Graph I is the derivative of graph II. The two zeros in graph I near $x = 3$ and $x = 5$ match the behavior of the local maximum near $x = 3$ and the local minimum near $x = 5$ in graph II.

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84.

Graph of f''

The graph of f'' , the second derivative of the function f , is shown above on the interval $0 \leq x \leq 6$. Which of the following could be the graph of f ?

**Answer C**

Correct. The graph of f'' is negative until $x = 1$, then positive until $x = 4$, then negative again. This means that the graph of f should be concave down, then concave up, then concave down, with two points of inflection at $x = 1$ and

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$x = 4$. This graph demonstrates that behavior and could therefore be the graph of f .